VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"Jnana Sangama", Belagavi-590018

A Project Report
On

"VOTING APPLICATION USING BLOCKCHAIN TECHNOLOGY"

SUBMITTED IN PARTIAL FULFILLMENT FOR THE AWARD OF DEGREE

BACHELOR OF ENGINEERING
IN
INFORMATION SCIENCE AND ENGINEERING

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2021 - 2022

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ACKNOWLEDGEMENT

We would like to express our profound grateful to His Divine Soul **Padmabhushan Sri Sri Sri Dr. Balagangadharanatha MahaSwamiji** and His Holiness **Jagadguru Sri Sri Sri Dr. Nirmalanandanatha MahaSwamiji** for providing us an opportunity to complete our academics in this esteemed institution.

We would also like to express our profound thanks to **Revered Sri Sri Dr. Prakashnath Swamiji**, **Managing Director**, SJB Institute of Technology, for his continuous support in providing amenities to carry out this project in this admired institution.

We express our gratitude to **Dr. K V Mahendra Prashanth**, **Principal**, SJB Institute of Technology, for providing us an excellent facilities and academic ambience; which have helped us in satisfactory completion of project work.

We extend our sincere thanks to **Dr. Rekha B**, **Professor & Head**, Department of Information Science and Engineering; for providing us an invaluable support throughout the period of our project work.

We wish to express our heartfelt gratitude to our guide**, Mr. Ranjith J**, **Assistant Professor,** Department of Information Science and Engineering for his valuable guidance, suggestions and cheerful encouragement during the entire period of our project work.

We also express our truthful thanks to our Project Coordinator, **Mr. Ranjith J, Assistant Professor,** Department of Information Science and Engineering, for his valuable support.

Finally, we take this opportunity to extend our earnest gratitude and respect to our parents, Teaching & Non-teaching staff of the department, and all our friends, who have directly or indirectly supported us during the period of our project work.

Regards,

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ABSTRACT

A widespread mistrust towards the traditional voting system has made voting in any country very critical. People have seen their fundamental rights being violated. Other digital voting systems have been challenged due to a lack of transparency. This makes it very difficult for the government to gain voters trust. The reason behind the failure of the traditional and current digital voting system is that it can be easily exploited. The primary objective is to resolve problems of the traditional and digital voting system, which include any kind of mishap or injustice during the process of voting.

Blockchain technology can be used in the voting system to provide a fair voting system and reduce injustice. The physical voting systems have many flaws in it as well as the digital voting systems are not perfect which appraises the need for a solution to secure the democratic rights of the people. Our project presents a framework to digitally conduct the voting activity based on the modern technology blockchain to provide maximum transparency of the system to build a trustful relationship between voters and election authorities. Our project eliminates the need of any physical polling stations. Smart contracts are used provide a secure connection between the user and the network while executing a transaction.

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CHAPTER 1

INTRODUCTION

1.1 Elections

In democracy based governance, elections play a key role in determining the government. If elections get rigged, then the entire nation will be under threat. The people will lose faith in the democratic system. According to the Election Commission of India, 814.5 million people were eligible to vote in general election conducted in 2014 and the average election turnout was around 66.38%. In order to increase the voting percentage, online secured voting system with enhanced security mechanism that prevents fraudulent votes and other malpractices is needed. Not only in India, other large democratic countries like USA, UK etc also suffer from malpractices during elections. In USA, many people think that voting systems at certain locations are rigged. This in turn induces distrust towards the democratic system. If there has to be a good democratic government, the elections held must be highly secured so that the people of the nation will have a greater trust on the system and will come forward to exercise their voting rights. [3]

1.2 Overview of Elections in India

India conducts elections to the Lok Sabha and Vidhan Sabha (Assembly) every five years to establish legal governance that is in compliance with the Constitution of India framed. The terms of all elected legislators are set to expire in five years. Either on the same day or within a few days, elections are held simultaneously in all constituencies. This is referred to as a General Election. Sometimes a single constituency election is called to fill a vacancy caused by a member's death or resignation. A "by-election" is a term for this type of election.

Following are types of elections seen in India:

a) Parliamentary general elections (Lok Sabha):

Members of the Lok Sabha (House of the People), or lower house of India's Parliament, are elected by all adult citizens of India from a list of candidates running for their particular seats. Every adult Indian citizen is only permitted to vote in the constituency in which they reside. "Members of Parliament," refers to the candidates who win the Lok Sabha elections and hold their seats for five years or until the President, on the advice of the council of ministers, dissolves the body. The house meets in the Sansad Bhavan, that is, Lok Sabha in New Delhi to discuss issues related to welfare of the country such as introduction of new laws, repeal of current laws, and improvement of existing laws that affect all Indian citizens. Once every five years, 543 members of the Lok Sabha are elected.

b) State Assembly (Vidhan Sabha) Elections:

Members of the State Legislative Assembly are elected from a field of candidates running in their districts by popular vote. Every adult Indian citizen is only permitted to vote in the constituency in which they reside. Candidates who win seats in the State Legislative Assemblies are referred to as "Members of Legislative Assembly" (MLA) and serve for five years or until the Governor dissolves the body. The house meets in each state to discuss issues such as the development of new laws, the repeal or improvement of existing laws that impact all citizens in that state.

c) Rajya Sabha (Upper House) Elections:

The Rajya Sabha, commonly known as the Council of States, is Indian Parliament's upper house. Candidates are chosen by Members of Legislative Assemblies rather than citizens, and up to 12 can be nominated by the President of India for contributions to art, literature, science, and social services. Members of the Rajya Sabha have a six-year term, with one-third of the body being up for re-election every two years. Before a bill becomes an act, the Rajya Sabha serves as a second-level review body. The Vice President of India serves as the Rajya Sabha's ex-officio Chairman, presiding over its proceedings. Legislative proposals (creating new laws, repealing or adding additional conditions to existing laws) are presented in the form of a bill to either house of Parliament.

Election Campaign:

The term "election campaign" refers to the promotion of the candidates' policies, offers, and promises to voters, which they intend to keep if they are elected. Voters can then choose who they want to vote for. They cast their ballots for the candidate whose policies they support. Between the announcements of the final list of candidates and the date of polling in India, election campaigning lasts two weeks. Candidates contact their voters, political leaders, speak in election meetings, and political parties mobilize their followers during this time. This is also the time of year when election-related stories and debates dominate television news and publications. Political parties begin campaigning for elections months before they take place. Political parties strive to focus public attention on a few major issues during election campaigns. They want to draw the public's attention to the issue so that they can vote on it.

Democratic Election in India:

Many variables contribute to the democratic nature of Indian elections. These are:

a) Independent Election Commission:

Elections in our country are overseen by an independent body called Election Commission of India (ECI). It possesses the same degree of autonomy as the judiciary. The Chief Election Commissioner (CEC) is appointed by the President of India. CEC is not accountable to the President or the government once appointed.

The Election Commission has the following powers:

- Makes all decisions related to conduct and control of elections, spanning from the announcement of elections till the declaration of results which marks the end of election activity.
- It is in charge of enforcing the code of conduct and punishes any candidate who violates the rules.
- It has the authority to order the government to observe certain standards throughout the election.
- These standards prohibit the use and abuse of government power in order to improve the administration's chances of winning elections or transferring some government employees.
- The Election Commission is in charge of government officers on election duty.
- If election officials believe that polling in some booths or possibly an entire constituency was unfair, they request a rerun.

b) Popular Participation:

Another option to assess the quality of the electoral process is to use this method. People will not continue to participate in the electoral process if it is not free and fair. Voter turnout is a common way of measuring people's election participation. The percentage of eligible voters who actually vote is known as turnout. In India, voter turnout has either remained constant or increased during the last 50 years. In India, the poor, illiterate, and downtrodden vote in greater numbers than the rich and privileged. India still follows the traditional voting system in which we use Electronic Voting Machine (EVMs). These need the deployments of the security personnel in the election centres; people need to travel and questions about the integrity of the election rises too. We can overcome this by implementing e-voting using block chain.

1.3 Traditional Voting System

In the traditional system of voting, all citizens above the age of 18 and 18 are eligible to participate in the voting activity. Before voting, the voter has to register himself with the election authorities after which, a verification check is made against certain conditions and then the voter ID is issued to the voter. On the day of the election the voter must verify his voter ID with the officials, only then he will be permitted to cast his vote. This requires the voter to travel long or short distances to the designated polling station. After the voting activity is concluded the EVMs are packed and transported to the strong room. To provide security to the EVMs, we can see heavy deployment of security personnel. Also, during the election process security deployment is made to ensure that things don't go in a wrong manner. Finally, the results are announced after 2 or 3 days. In addition to the security personnel, large numbers of staffs are required to ensure election process takes place smoothly without any hassle. Also, resources like VVPAP paper is given to the voter to bring trust in voters that their voting integrity is secured and non-erasable ink is put on finger to ensure that voters don't vote twice.

The detailed working of a traditional voting system can be seen in figure 1.1

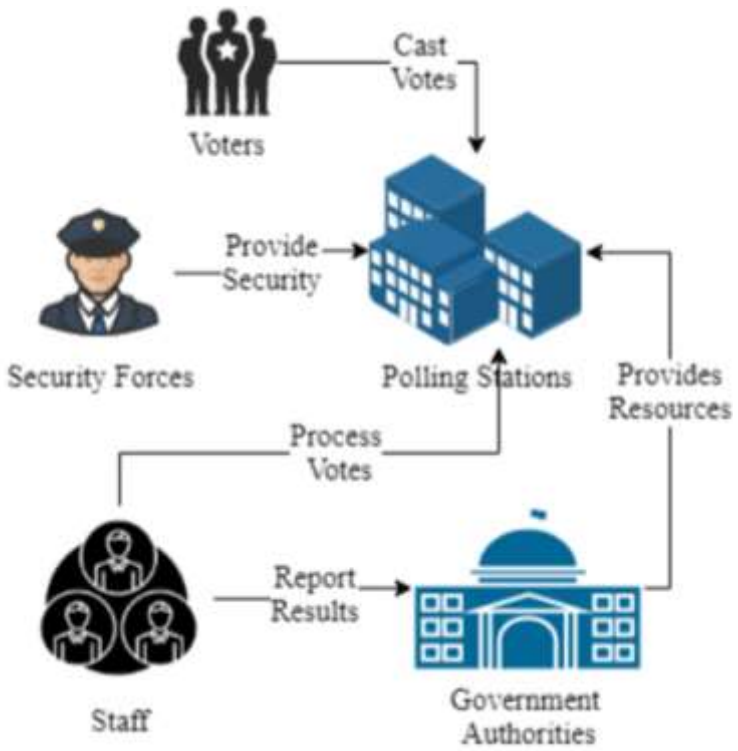

Fig 1.1: Traditional Voting System

1.4 E-Voting Background

Electronic voting has been an area of research for many years by using computing machines and equipment for casting votes and producing high quality and precise results in accordance with the sentiments of the participating voters. Various attempts have been adopted in practice to support election process. Initially computer counting system allowed the voter to cast vote on papers. Later on, those cards went through the process of scanning and tallying at every polling cell on a central server. Direct Recording Electronic (DRE) voting systems were put in place later, which were admired and acknowledged greatly by the voter's in-spite of the resistance from computer scientists. If the voting system is well understood by the voters, the system's usability can be increased remarkably. DRE systems in particular have gathered a lot of successes in bringing the voters to use this technology. These systems work more or less in the same way as any conventional election system does. In the case of DRE, a voter begins his journey by going to their polling place and gets their token to vote where he utilizes his token at the voting terminal to vote for his candidate. When the candidate selection procedure is completed, DRE systems present the final selection to the voter before actually casting it (in case if the voter wants to change his opinion) and after the final selection, the ballot casting is completed. More recently, distributed ledger technologies such as blockchain have been used to achieve e-voting systems primarily due to their advantages in terms of end-to-end verifiability. With properties such as anonymity, privacy protection and non-repudiation, blockchain is a very attractive alternative to contemporary e-voting systems. The model proposed by us attempts to leverage these properties of blockchain to achieve an efficient e-voting system.

1.5 E-Voting Requirements

The requirements of an E-Voting System are:

a) Privacy - Keeping an individual's vote secret:

The system leverages cryptographic properties of blockchain to achieve privacy of a voter. More specifically, as voter is registered into the system, a voter hash is generated by blockchain which is the unique identifier of a voter into the blockchain, and is protected from misuse due to collision resistance property of the cryptographic hash. Due to this, the traceability of a vote is also non-trivial thereby protecting the voter when under duress.

b) Eligibility - Allowing only registered voters to vote, with each such voter votes only once:

All eligible users are required to register using unique identifiers such as government-issued documents to assert their eligibility. In addition to this, our system implements a check to assert that only authorized voters can access the system.

c) Receipt Freeness - Voters be unable to prove to third party to whom they voted:

The proposed system enables a voter to vote as per their choice and creates a cryptographic hash for each such event (transaction). This is important to achieve verifiability i.e. to verify if a certain vote was included in the count. However, possession of this hash does not allow extracting information about the way voter has voted.

d) Convenience - Voters must be able to vote easily and everyone eligible must be able to vote:

The system implements a user friendly web based interface with the voting process requiring minimal input from the user. Furthermore, the overall process is integrated which enables the user to interact with it in a seamless manner.

e) Verifiability - The ability to trust the vote tallying process:

Upon casting their vote successfully, a user is provided with their unique transaction ID in the form of a cryptographic hash. A user can use this transaction ID to track if their vote was included in the tallying process. However, this process does not enable a user to view how they voted which has been adopted to mitigate threats when under duress.

The analysis presented above highlights the performance of the proposed system with respect to the specific requirements of e-voting. It also highlights the significance of defining characteristics of blockchain and its role in achieving the cornerstones of an efficient e-voting system. Therefore, we believe that this system makes significant contribution to the existing knowledge with respect to the application of blockchain technology to achieve a secure digital voting system.

1.6 Motivation

Our main motivation in this project is to provide a secure voting environment and show that a reliable e-voting scheme is possible using blockchain. Because, when e-voting is available for everyone who has a computer, or a mobile phone, every single administrative decision can be made by people and members; or at least people's opinion will be more public and more accessible by politicians and managers. This will eventually lead humanity to the true democracy. It's important for us since elections can easily be corrupted or manipulated especially in small towns, and even in bigger cities located in corrupt countries. Plus, large-scale traditional elections are very expensive in the long term, especially if there are hundreds of geographically distributed vote centers and millions of voters. Also, the voters (mainly for members of organizations) might be on vacation, on a business trip or far away for any other reason, which will make impossible for that particular voter to attend the election and may lower the overall voting percentage. E-voting will be able solve these problems, if implemented carefully.

1.7 Challenges

Today, when elections are being conducted, the voting percentage turnover at the end of the voting activity is seen to be very less. Also, the key treat to the current voting system is the security lapse where some election officials may be threatened to do work as directed by the standing candidate so that the votes may fall in favour of him. There may also be chances of manipulation of hardware so as to get maximum votes. Although, these challenges are faced in the current scenario, yet no steps are taken to implement a voting system based on blockchain technology. In our project, we aim to build a blockchain based reliable and secure voting system through which the challenges faced in the current system can be slightly reduced.

1.8 Objectives

The objectives of the application system proposed by us is to

- Design a tamper proof voting system so as to guarantee that the voters voting integrity is maintained.
- Ensure voters vote integrity.
- Ensure fast availability of voting results.

CHAPTER 2

LITERATURE SURVEY

The objective of carrying out literature survey before implementing any project is to get an idea of how the project can be implemented and to understand various technologies used to build a project. So it is of utmost necessary to find the most recent and promising research papers related to our project topic "Voting Application using Blockchain Technology". The papers referred to have their publication dates in range between 2016 and 2022. The details about the contributions made by each author are given below.

2.1 Research Papers

Some of the papers referred are as follows:

Secure and Transparent Election System for India using Block chain Technology – 2019 [12]

Here, the model system proposed by the authors use Ethereum to support smart contracts. The authors have provided an easy interface to allow voters to cast their votes. On the voter side, a browser-based interface is provided to cast votes making it very user-friendly interface. At the time of voting, each user is informed about the Ethereum wallets that are associated with their corresponding Voter ID for the election polling. So, when the voter is directed to the portal, he has to register there with their Aadhar card and select the constituency from the drop-down menu that is provided to them as the system is online. Once the user is registered to the system, a hash code is provided in which his identity is encoded. So, at the time of voting voter without any hassle of A voter need not go back to their constituency to cast their vote. In the block chain network, the first block in each block-chain called the genesis block which would contain all the relevant information about the election candidate standing for elections. Each block except the genesis block, in the blockchain(s) would contain one transaction, that is, one vote. Hence, the length of a candidate's blockchain (neglecting the genesis block) will directly give the number of votes the candidate received. The miners (or the validating nodes) in this system will be the officials of the Election commission. Their work is to validate each transaction (that is, vote) on various block chains. Each voter is provided with a wallet on Ethereum for a particular election by the Election Commission. Because transactions require some gas (or some funds), each wallet of a voter will have just enough gas required to do exactly one transaction (that is, cast exactly one vote). Only the Election Commission can add gas to the Voter wallets. This ensures that only one vote is cast per voter and at the same time nullifies the possibilities of vote duplication. Each transaction is secured with public-private key cryptography where the vote (or the transaction)

will be first encrypted with the private key of the voter's wallet. This is called signing the transaction. Then, this would be further encrypted with the public key of the Election Commission's wallet. Next, when this transaction is mined, the miners will first decrypt the transaction using the public key of the Election Commission's wallet. Then, this will be further decrypted using the public key of the corresponding voter wallet's public key.

Votereum: An Ethereum-based E-voting system – 2019 [13]

Here, the authors developed a system called votereum which is an Ethereum based e-voting system built on the principles of block chain technology of Ethereum platform. The proposed system used a web interface developed using angular web development technology. The system also uses an application server and a Restful API service. Authentication will authenticate access to the API using JWT token. After login to the system, user will receive a JWT token which then will be used to access to the API within their permission level like cast vote, set up a new ballot, etc. For blockchain related request, Authentication server will communicate with OBR (Online Ballot Regulator) through AMPQ protocol. The OBR is responsible for handle request to Ethereum from AS in the request queue. It will execute the functions in the smart contract and wait for the result from the network. Because the response time often takes from three seconds to more than one minute, so the AS and OBR will use AMPQ protocol to send and receive data from blockchain. For example, AS request create a new ballot, it will send to the AMPQ queue and subscribe to this queue, the AMPQ protocol is wrapped up by RabbitMQ – the message broker. A local database is also equipped at the backend to store information of citizen and candidate. The citizen data includes user system account information, Ethereum address and encrypted password (for cast vote), and detail information of that citizen. The candidate data includes their name, citizen ID and some related information about them. To interact with the network, an Ethereum node is required to be running in our server which in this case is inside the OBR. This acts as a gateway providing interfaces to interact with the network.

A Novel P2P based System with Blockchain for Secured Voting Scheme – 2019 [14]

Here, in the voting system proposed by the authors in this paper, they have developed a web application responsible for creating and managing the voting events. The Admin stores the information such as Aadhar number, constituency and the details necessary for the candidate through a web application. These details are then carried to the server and maintained. The voting event takes place through a web application, where the voter is allowed to cast their vote from anywhere i.e, they can visit to the nearby booth and poll their vote. The server will authenticate each user by Aadhar number. In order for the user to login into the application corresponding OTP will be generated for the registered mobile number. The user must enter the Aadhar number and the OTP based on which the corresponding contestant are notified to the user. The system makes use of an authentication server is

an application that provides network service and used to authenticate the credentials such as Aadhar number of the user. It consists of a backend database to store the information of all the citizen of the nation. It also makes use of an arbitration server is responsible for tracking the entire application server and it acts as an intermediary of blockchain based balloting system and a user. It verifies the registered user during the process of voting using the authentication server. Arbitration server is used to send the users' vote to a blockchain node. This in return sends the user the block number through which the user can verify their vote. Once the vote is casted by the user, double encryption takes place on the data using AES and RSA algorithms.

Decentralised Applications Using Ethereum Blockchain – 2019 [15]

Here, an Ethereum blockchain based system is proposed by the authors to perform voting activity. Ethereum provides a wide range of services and solutions with its readily available development tools and smart contracts. A smart contract is a self-executing program that runs on the blockchain. In order for it to get compiled and deployed on the blockchain, each node on the network then executes the contract in exchange for some ether. The currency required to execute a contract is called gas amount and varies from contract to contract. Solidity is used as a programming language that runs directly on the blockchain. The smart contracts are written in solidity. Contracts are executed by all the nodes and the updated information is shared amongst other nodes after a regular interval. These contracts should be validated by at least 2 nodes to become activated. MetaMask is an extension added to Chrome that connects to the local Ethereum network which is used for this application with the personal account and the ability to interact with smart contract application which has already been created. The framework of our choice is Ganache which is powered by truffle and has 10 accounts with 100ETH each to work with. Voting on the blockchain uses a fraction of our ether as gas amount. Once the gas is paid, the transaction occurs and all the other nodes get to know about this transaction as well so the voting occurs in real time and becomes difficult to tamper with. Since a user can vote only once, after voting, he/she doesn't get the interface to vote again but instead they get to see the results of the election.

A Privacy-Aware Digital Voting System employing Blockchain and Smart Contracts – 2020 [18]

In the voting system proposed by the authors of this paper, they suggested that voters can cast their votes from anywhere via smart devices. It is not necessary to stand in the voting centers in long lines and expect some sort of abuse from the elections. People without smartphones can still cast their ballots from the specified polling station. Before voting, the voter should register by providing details such as NID, name, security key and mobile number. A hash value will generate using SHA-256 hash function from these inputs. Generated hash value is stored by Smart Contract in blockchain. A key algorithm called elliptical curve cryptography algorithm is used for the generation of a key pair of

public and private key. Public keys are used in the blockchain network to identify the voters. A private key is forwarded to the number of voter's mobile. Using this private key, voters can sign in blockchain wallet and cast their votes. The Candidate Registration process is same as like voter registration. The voter authentication process is done by Smart contract. First, the Voter must sign in their wallet using their private key, Then submit the credentials for authentication by admin. Smart contract generates the hash value from the credentials to match the hash with existing hash values in the blockchain. If both hash value matches then it send the verification result of voter. A vote is represented as a transaction in the blockchain. After performing the authentication process voter can cast vote. A vote id (VID) will be generated at client site. The casted vote and VID will be encrypted with the public key of election commission. During the counting process Election commissioner inputs the private key. Each encrypted vote will be decrypted from voting chain and the smart contract will send the casted vote to the each choosen candidate public key. The vote is counted immediately by smart contract and then publishes the results. It is the same as calculating the total vote held by a specific address.

Decentralized E-voting system based Smart-Contract using Blockchain Technology – 2021 [19]

Here, the authors have developed an e-voting application based on decentralised architecture - blockchain technology. Before using this system the voter and the candidate registration process has to done in advance. Identity must be verified before creating accounts. After verifying Identity, the authorized person must authenticate eligible users by verifying a coin or token. In this application, the code writes to the decentralized application, by using smart contract protocol in ethereum blockchain. The Ethereum allows to write a code and executes it on Ethereum virtual machine smart contracts. The application has made use of HTML, CSS and JS to build the front end to provide easy user interface. Truffle offers a set-up of instruments, which encourage programming to compose Smart Contracts with the Solidity programming language. It additionally gives a structure to test Smart Contracts. In this application Ganache is used as a local storage for development of e-voting decentralized. MetaMask is a google chrome extension. MetaMask connects to the local Ethereum network which is used for this application with the personal account and the ability to interact with smart contract application which has already been created.

Blockchain based Voting system in Local Network– 2021 [20]

Here, the voting system proposed by the authors is built on block chain technology. The voters before participating in the election are given with a public address and a private key (both should not be revealed to anyone) that can be used for voter authentication. Also, each voters account is filled with 100 ethers by default to carry out transactions. Once the voter's credentials are entered as prompted by the system, upon verification is allowed to cast their vote to the candidate in favour. Only the admin or

the person who initiates the election is eligible to add candidates by using the front end application. To cast a vote, a function call is made to the deployed smart contract with the candidate id. The transaction of the block is successful only when the transaction id conformed by the miner and acknowledged by all the miners. To create the Ethereum blockchain network, the authors have used the tool Ganache in the local machine and the smart contract is deployed in the network using the Truffle framework. Ganache provides 10 accounts with 100 fake ethers which can be used to test the working of the developed application. When the smart contract is triggered Migrations are used to update the blockchain. For each migration, certain amount of ether is spent from the account which is used to deploy the smart contract in the blockchain network. When the user clicks the vote button, metamask will ask the user for confirmation. On confirming, the transaction is added to the Pending transaction pool. After the Votes are casted by the voters, it is stowed as immutable and tamper-proof. This addresses the security issues with the current EVM system. Though it provides transparency, as the transactions are visible to everyone, it conserves voter's confidentiality and secrecy. It helps in announcing the result fast. The voting results are publicly auditable.

Towards A Privacy-Preserving Voting System through Blockchain Technology – 2019 [22]

Here, the authors have proposed a digital voting system that uses blockchain technology. The system works by first authenticating the voter. If the voter is a verified voter, then he is eligible to take place in the voting activity else the voters are prompted to first register themself with the voting management system. Only verified voters can cast their votes. After the votes are casted by verified voters, they are then confirmed by the polling agents. If the votes are confirmed they are added to the blockchain else the votes are passed on to another person to check for authenticity. If error in the vote is seen then that vote is cancelled else it is added to the block chain. Finally, from the blockchain votes are retrieved and audited and results are published. The authors have also shown keen interest in improving the systems security. To maintain voters vote integrity, merkle tree introduced by Ethereum to allow a compact and efficient verifiable proof which ensures that a transaction is included in a block. Every block in Ethereum contains three Merkle trees for three kinds of objects; transaction, receipts and state. In this data structure, hashes of child nodes are combined together into the parent node's header. This technique of combining the child nodes' headers and adding it to the header of the parent node continues iteratively until it reaches the final node, right at the top, the root node. Thus, the root node will contain information about all of the nodes present in the tree. Hash of all the transactions in a block is stored in the Merkle tree. Voter's authentication is made by the Voter Authentication Unit (VAU). The proposed system also offers anonymity and privacy to the voters.

2.2 Existing System

The following are the drawbacks found in the existing system:

- Time consuming as it requires voters to travel to designated polling station and stand in long queues to cast their votes.
- The voters are required to file a manual application to get their voter id.
- The voting percentage turnover is less as some people may not be able to cast their vote since they may be out of station on personal reasons.
- Voting results are delayed.

2.3 Proposed System

In the proposed system we aim to

- Ensure that the voting percentage turnover is high by making our system portable so that people from any corner of the world are able to vote.
- Provide voting integrity to voters vote and make sure that no manipulation takes place within Voting System.
- Make voting results available faster.

CHAPTER 3

SYSTEM REQUIREMENTS SPECIFICATION

System requirements specification establishes the basis for an agreement between customers and contractors or suppliers on how the software product should function. SRS is a rigorous assessment of requirements before the more specific system design stages, and its goal is to reduce later redesign. It should also provide a realistic basis for estimating product costs, risks, and schedules. Used appropriately, software requirements specifications can help prevent software project failure.

A Software Requirements Specification is a detailed description of a software system to be developed with its functional and non-functional requirements. The SRS is developed based on the agreement between customer and contractors. It may include the use cases of how the user is going to interact with software system. The software requirement specification document consistent of all necessary requirements that is required for project development. To develop the software system, we should have a clear understanding of the Software system.

3.1 Hardware Requirements

- System : Intel i5 2.4 GHz
- Hard Disk : 500GB
- RAM : 4/8GB

3.2 Software Requirements

- Operating System: Windows 7
- Browser: Google Chrome
- IDE: Visual Studio Code
- Software Tools: Ethereum Block Chain, Ganache, Truffle, Web3js, Metamask
- Coding Language: Solidity, React JS

3.3 Tools Used

Visual Studio IDE

An integrated development environment (IDE), also known as integrated design environment and integrated debugging environment, is a type of computer software that assists computer programmers to develop software. The Visual Studio IDE consists of several sections, or tools, that the developer

uses while programming. The code is written in modules, or files, that are either connected to an object or called specifically when needed.

React JS

React is a free and open-source front-end JavaScript library for building user interfaces based on UI components. It is maintained by Meta (formerly Facebook) and a community of individual developers and companies. React can be used as a base in the development of single-page, mobile, or server-rendered applications with frameworks like Next.js. However, React is only concerned with state management and rendering that state to the DOM, so creating React applications usually requires the use of additional libraries for routing, as well as certain client-side functionality.

Ethereum Blockchain

Ethereum is a decentralized global software platform powered by blockchain technology. It is most commonly known for its native cryptocurrency, ether, or ETH. Ethereum can be used by anyone to create any secured digital technology. Ethereum is designed to be scalable, programmable, secure, and decentralized. Blockchain is an information management technology that uses a distributed processing architecture to enhance the resiliency of data transactions from unauthorised changes, hacks, and corruptions. Ethereum offers an extremely flexible platform on which to build decentralized applications using the native Solidity scripting language and Ethereum Virtual Machine. Decentralized application developers who deploy smart contracts on Ethereum benefit from the rich ecosystem of developer tooling and established best practices. Ethereum encourages developers to deploy their applications on the network, which further reinforces Ethereum as the primary home for decentralized applications like DeFi and NFTs.

Solidity

Solidity is an object-oriented, high-level language for implementing smart contracts. Smart contracts are programs which govern the behaviour of accounts within the Ethereum state. Solidity is a curly-bracket language designed to target the Ethereum Virtual Machine (EVM). It is influenced by C++, Python and JavaScript. Solidity is statically typed, supports inheritance, libraries and complex user-defined types among other features. With Solidity you can create contracts for uses such as voting, crowd funding, blind auctions, and multi-signature wallets. Solidity supports complex member variables for contracts, including arbitrarily hierarchical mappings and structs. Solidity contracts support inheritance, including multiple inheritances with C3 linearization.

Metamask

MetaMask is a software cryptocurrency wallet used to interact with the Ethereum blockchain. It allows users to access their Ethereum wallet through a browser extension or mobile app, which can

then be used to interact with decentralized applications. MetaMask allows users to store and manage account keys, broadcast transactions, send and receive Ethereum-based cryptocurrencies and tokens, and securely connect to decentralized applications through a compatible web browser or the mobile app's built-in browser. Developers achieve a connection between Metamask and their decentralized applications by using a JavaScript plugin such as Web3js or Ethers to define interactions between Metamask and Smart Contracts.

Ganache

Ganache is used for setting up a personal Ethereum Blockchain for testing your Solidity contracts. It provides more features when compared to Remix. Ganache is part of the Truffle Suite ecosystem. Specifically, the Truffle Suite consists of Ganache and an additional pair of tools; Truffle and Drizzle. Truffle is a development environment and testing framework using the EMV (Ethereum Virtual Machine); meanwhile, Drizzle is a collection of frontend libraries. Ganache allows you to set up your blockchain, which in turn provides a safe testing environment for your dApps and smart contracts.

Truffle

Truffle is a world-class development environment, testing framework and asset pipeline for blockchain using the Ethereum Virtual Machine (EVM), aiming to make life as a developer easier. Truffle is widely considered the most popular tool for blockchain application development. Paired with Ganache, a personal blockchain, and Drizzle, a front-end dApp development kit, the full Truffle suite of tools promises to be an end-to-end dApp development platform. Truffle helps in easy deployment of smart contracts and communicates with their underlying state without heavy client-side programming.

CHAPTER 4

SYSTEM DESIGN

System design is the process of defining the architecture, components, modules, interfaces, and data for a system to satisfy specified requirements. System design is one of the important phases of software development process. The purpose of the design is to plan the solution of a problem specified by the requirement documentation. In other words, the first step in the solution to the problem is the design of the project. The design concepts provide the software designer with a foundation from which more sophisticated methods can be applied. Software design is a process of problem-solving and planning for a software solution.

After the purpose and specifications of software are determined, software developers will design or employ designers to develop a plan for a solution. It includes low-level components and algorithm implementation issues as well as the architectural view. Object-oriented analysis and design methods are becoming the most widely used methods for computer system design. It is widely used for modelling software systems and is increasingly used for high designing non-software systems and organization. System design is the process of defining the architecture, components, modules, interface and data for a system to satisfy specified requirements. System design could see it has the application of systems theory to product development.

4.1 Design Overview

The increasing importance of software running on generic platforms has enhanced the discipline of software engineering. Object-oriented analysis and design methods are becoming the most widely used methods for computer systems design. The UML has become the standard language in object-oriented analysis and design. It is widely used for modelling software systems and is increasingly used for high designing non-software systems and organizations.

System design is one of the most important phases of software development process. The purpose of the design is to plan the solution of a problem specified by the requirement documentation. In other words, the first step in the solution to the problem is the design of the project.

The design will contain the specification of all those modules their interaction with the other modules and desired output from each module. The output of design process is a description of software architecture. The design of our proposed model is shown in figure 4.1.

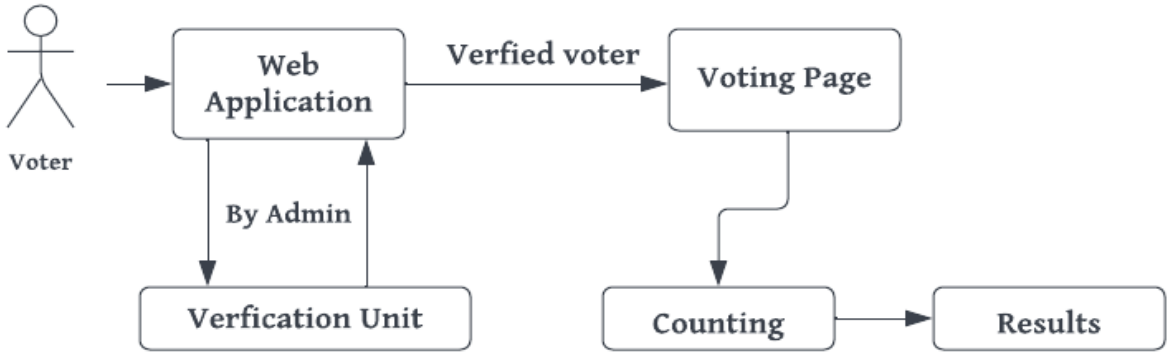

Fig 4.1: Proposed Model Design

Here, in our model the voter uses the web application of e-voting system either to register himself for the voting activity or to cast his precious vote to the candidate in favour. Before casting the vote, the voter has to get verified by the admin (person who initiates the voting activity) after which the voter can participate in the voting activity. After the voting activity is concluded, the votes are secured in the block chain and the admin can count the votes with the help of block chain and announce the results.

4.2 System Architecture

A system architecture or systems architecture is the conceptual model that defines the structure, behaviour, and more views of a system. An architecture description is a formal description and representation of a system, organized in a way that supports reasoning about the structures and behaviours of the system. A system architecture can consist of system components and the subsystems developed, that will work together to implement the overall system. There have been efforts to formalize languages to describe system architecture; collectively these are called Architecture Description Languages (ADLs). Figure 4.2 shows the architecture of our proposed e-voting system.

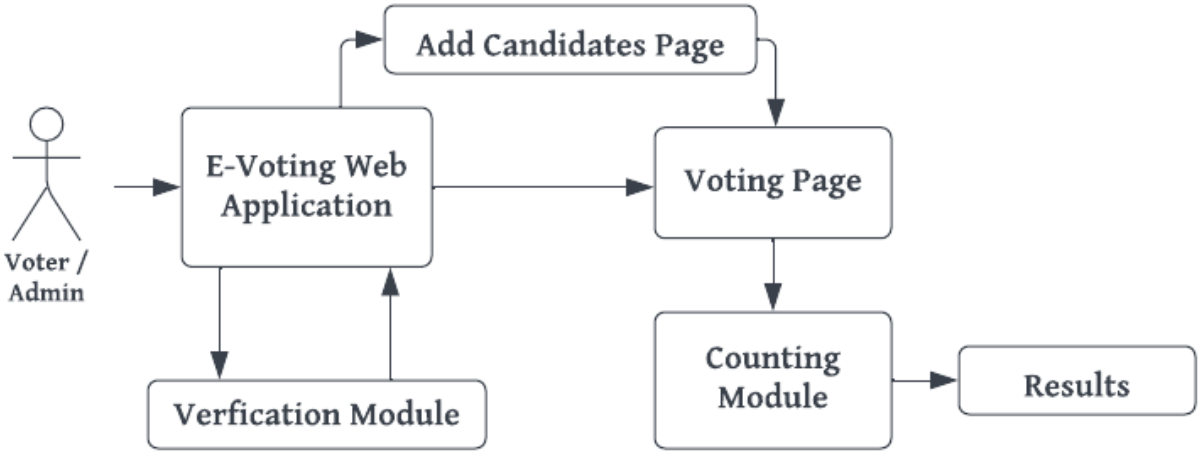

Fig 4.2: E-Voting System Architecture

4.3 Use Case Diagram

A use case diagram at its simplest is a representation of a user's interaction with the system that shows the relationship between the user and the different use cases in which the user is involved.

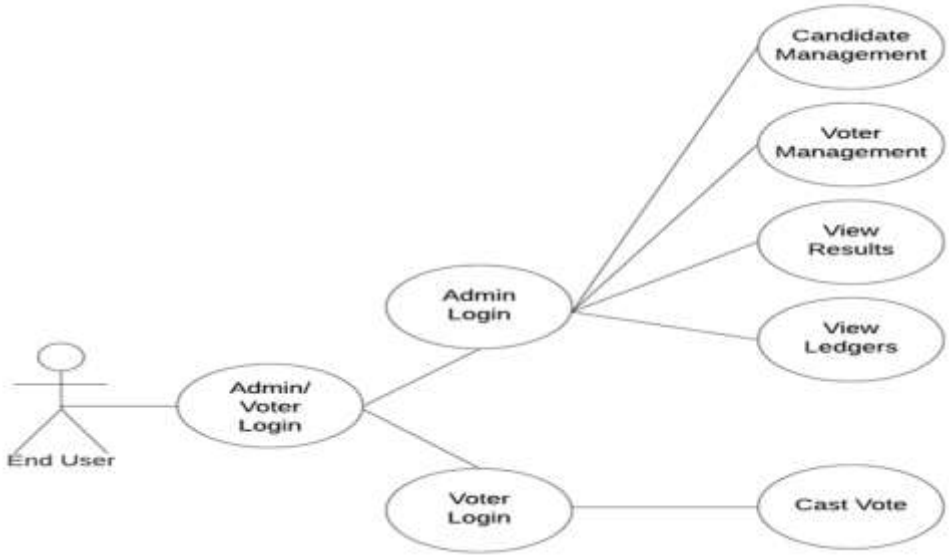

Fig 4.3: Use Case Diagram

4.4 Data Flow Diagram

A Data Flow Diagram (DFD) is a graphical representation of the "flow" of data through an information system, modelling its process aspects. A DFD is often used as a preliminary step to create an overview of the system without going into great detail, which can later be elaborated. DFDs can also be used for the visualization of data processing (structured design).

A DFD shows what kind of information will be input to and output from the system, how the data will advance through the system, and where the data will be stored. It does not show information about process timing or whether processes will operate in sequence or in parallel, unlike a traditional structured flowchart which focuses on control flow, or a UML activity workflow diagram, which presents both control and data, flows as a unified model.

Figure 4.3 shows the data flow diagram of our e-voting system.

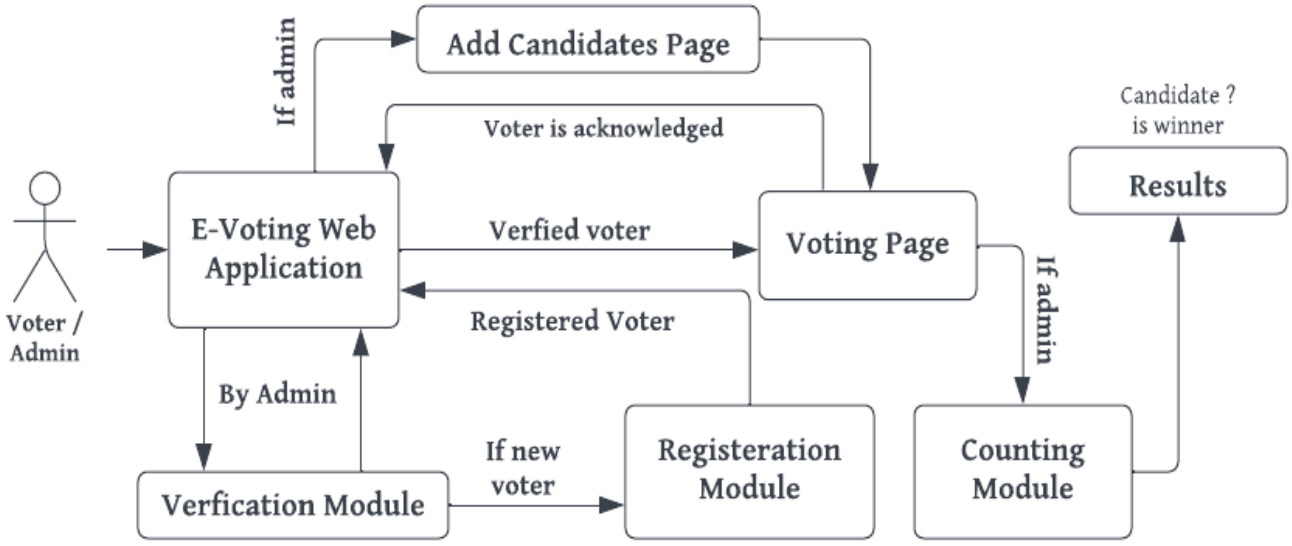

Fig 4.4: Data Flow Diagram of Proposed Model

Here, from the above figure both the voter as well as the admin can access the e-voting system web application. At the voter side, the voter can register with the system to participate in the voting activity by providing the required details. At the admin side, the candidates who wish to contest in the election can submit a manual application to the person who initiates the election or to the admin. The admin verifies the details given in the candidates' application. If all the details in the application are found to be appropriate then that candidate is added by the admin to the voting page. Also the admin performs voter authentication. If the voters are valid, then they are directed to the voting page to cast their votes else they are directed to the registration page. Once all the candidates are added and voters are found to be valid, the election is started by the admin and the voters can cast their votes. After the election is concluded, the admin ends the voting activity and the votes are secured in the blockchain. When the time has come to publish the results, the admin goes to the results pages to derive the results.

4.5 Sequence Diagram

A sequence diagram is a diagram that gives us the details about the interactions that takes place between the application and its users and in what order these interactions occur.

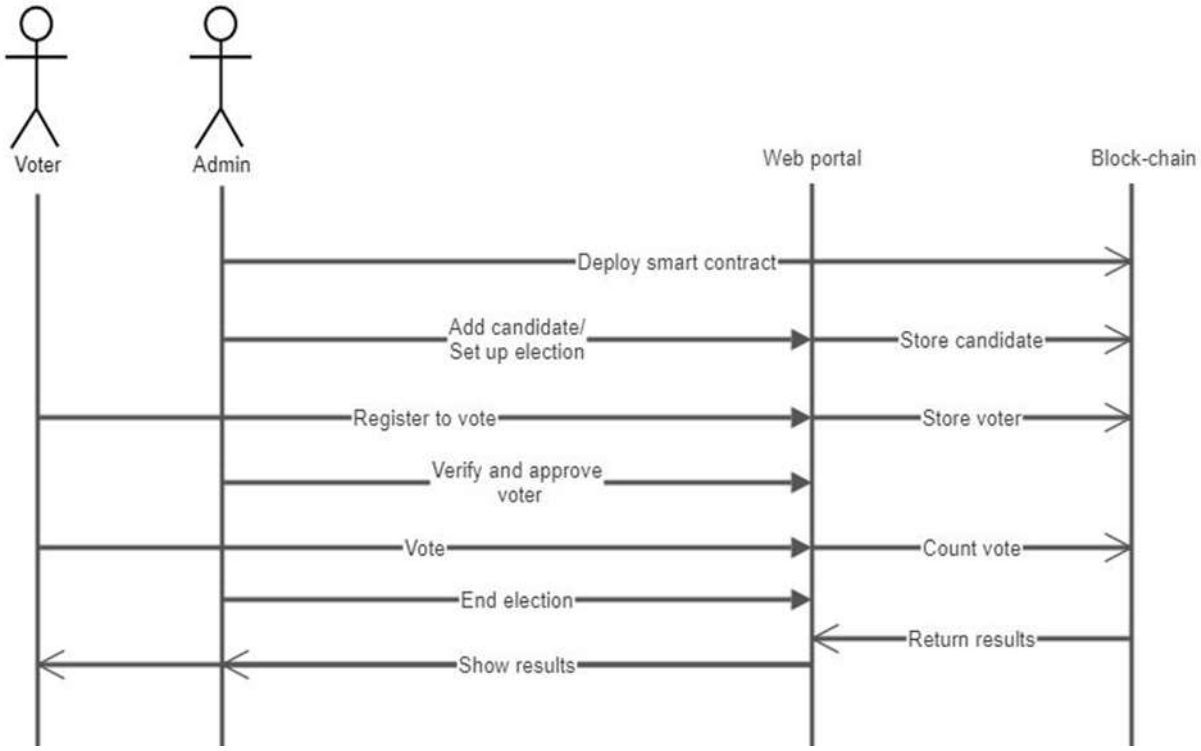

Fig 4.5: Sequence Diagram

CHAPTER 5

SYSTEM IMPLEMENTATION

Implementation is the process of defining how the system should be built, ensuring that it is operational and meets quality standards. It is a systematic and structured approach for effectively integrating a software-based service or component into the requirements of end users. System Implementation is the process of ensuring that the information system is operational and used, ensuring that it meets the quality standards. It uses the structure created during architectural design and the result of system analysis to construct system elements that meets the stakeholder requirements and system requirements developed in the early life cycle phases.

5.1 Modules Undertaken

The numbers of modules implemented in this project are five and are listed below as follow:

- Module 1: Adding Candidate
- Module 2: Voter Registration
- Module 3: Verification of Voters
- Module 4: Voting Process
- Module 5: Display Results

5.1.1 Module 1: Adding Candidate

This module provides a provision for the admin or the person who initiates the election to add candidates interested to stand in elections. The code snippet to implement this module is given below:

Code Snippet:

```
addCandidate = async () => {
    await this.state.ElectionInstance.methods
      .addCandidate(this.state.header, this.state.slogan)
      .send({ from: this.state.account, gas: 1000000 });
    window.location.reload();
  };
  render() {
    if (!this.state.web3) {
      return (
        <> {this.state.isAdmin?<NavbarAdmin/>:<Navbar/>}<center>Loading.</center></>
      );
    }
    if (!this.state.isAdmin) {
      return (
```

```
        <><Navbar /> <AdminOnly page="Add Candidate Page." /></>
      );
    }
    return (
      <>
        <NavbarAdmin />
        <div className="container-main">
          <h2>Add candidate</h2>
          <small>Total candidates: {this.state.candidateCount}</small>
          <div className="container-item">
            <form className="form">
              <label className={"label-ac"}>
                Name of candidate
                <input
                  className={"input-ac"}
                  type="text"
                  placeholder="Mr. ABC"
                  value={this.state.header}
                  onChange={this.updateHeader}
                />
              </label>
              <label className={"label-ac"}>
                Party name
                <input
                  className={"input-ac"}
                  type="text"
                  placeholder="ABC Party"
                  value={this.state.slogan}
                  onChange={this.updateSlogan}
                />
              </label>
              <button
                className="btn-add"
                disabled={this.state.header.length<3||this.state.header.length>21}
                onClick={this.addCandidate}
              >
                Add
              </button>
            </form>
          </div>
        </div>
        {loadAdded(this.state.candidates)}
      </>
    );
  }
```

5.1.2 Module 2: Voter Registration

This module provides a provision for the voters to register themselves in order to take part in the voting activity. The code snippet to implement this module is given below:

Code Snippet:

```
registerAsVoter = async () => {
    await this.state.ElectionInstance.methods
      .registerAsVoter(this.state.voterName, this.state.voterPhone)
      .send({ from: this.state.account, gas: 1000000 });
    window.location.reload();
  };
  render() {
    if (!this.state.web3) {
      return (
        <>
          {this.state.isAdmin ? <NavbarAdmin /> : <Navbar />}
          <center>Loading Web3, accounts, and contract...</center>
        </>
      );
    }
    return (
      <>
        {this.state.isAdmin ? <NavbarAdmin /> : <Navbar />}
        {!this.state.isElStarted && !this.state.isElEnded ? (
          <NotInit />
        ) : (
          <>
            <div className="container-item info">
              <p>Total registered voters: {this.state.voters.length}</p>
            </div>
            <div className="container-main">
              <h3>Registration</h3>
              <div className="container-item">
                <form>
                  <div className="div-li">
                    <label className={"label-r"}>
                      Account Address
                      <input
                        className={"input-r"}
                        type="text"
                        value={this.state.account}
                        style={{ width: "400px" }}
                      />{" "}
                    </label>
                  </div>
                  <div className="div-li">
                    <label className={"label-r"}>
                      Name
                      <input
                        className={"input-r"}
                        type="text"
                        placeholder="eg. Ava"
                        value={this.state.voterName}
                        onChange={this.updateVoterName}
                      />{" "}
                    </label>
                  </div>
```

```
                <div className="div-li">
                  <label className={"label-r"}>
                    Voter ID
                    <input
                      className={"input-r"}
                      type="text"
                      placeholder="IND-xxxx"
                      value={this.state.voterPhone}
                      onChange={this.updateVoterPhone}
                    />
                  </label>
                </div>
                <button
                  className="btn-add"
                  disabled={this.state.currentVoter.isVerified}
                  onClick={this.registerAsVoter}
                >
                  {this.state.currentVoter.isRegistered
                    ? "Update"
                    : "Register"}
                </button>
              </form>
            </div>
          </div>
          <div
            className="container-main"
            style={{
              borderTop: this.state.currentVoter.isRegistered
                ? null
                : "1px solid",
            }}
          >
            {loadCurrentVoter(
              this.state.currentVoter,
              this.state.currentVoter.isRegistered
            )}
          </div>
          {this.state.isAdmin ? (
            <div
              className="container-main"
              style={{ borderTop: "1px solid" }}
            >
              <small>TotalVoters: {this.state.voters.length}</small>
              {loadAllVoters(this.state.voters)}
            </div>
          ) : null}
        </>
      )}
    </>
  );
}
```

5.1.3 Module 3: Verification of Voters

This module makes sure that only eligible voters are able to take part in the voting activity. Thus this module allows the admin to approve only eligible voters. The code snippet to implement this module is given below:

Code Snippet:

```
renderUnverifiedVoters = (voter) => {
    const verifyVoter = async (verifiedStatus, address) => {
      await this.state.ElectionInstance.methods
        .verifyVoter(verifiedStatus, address)
        .send({ from: this.state.account, gas: 1000000 });
      window.location.reload();
    };
    return (
      <>
        {voter.isVerified ? (
          <div className="container-list success">
            <p style={{ margin: "7px 0px" }}>
              Account address: {voter.address}
            </p>
            <table>
              <tr>
                <th>Name</th>
                <th>Voter ID</th>
                <th>Voted</th>
              </tr>
              <tr>
                <td>{voter.name}</td>
                <td>{voter.phone}</td>
                <td>{voter.hasVoted ? "True" : "False"}</td>
              </tr>
            </table>
          </div>
        ) : null}
        <div
          className="container-list attention"
          style={{ display: voter.isVerified ? "none" : null }}
        >
          <table>
            <tr>
              <th>Account address</th>
              <td>{voter.address}</td>
            </tr>
            <tr>
              <th>Name</th>
              <td>{voter.name}</td>
            </tr>
            <tr>
              <th>Voter ID</th>
              <td>{voter.phone}</td>
            </tr>
```

```
            <tr>
              <th>Voted</th>
              <td>{voter.hasVoted ? "True" : "False"}</td>
            </tr>
            <tr>
              <th>Verified</th>
              <td>{voter.isVerified ? "True" : "False"}</td>
            </tr>
            <tr>
              <th>Registered</th>
              <td>{voter.isRegistered ? "True" : "False"}</td>
            </tr>
          </table>
          <div style={{}}>
            <button
              className="btn-verification approve"
              disabled={voter.isVerified}
              onClick={() => verifyVoter(true, voter.address)}
            >
              Approve
            </button>
          </div>
        </div>
      </>
    );
  };
```

5.1.4 Module 4: Voting Process

This module allows the voters to cast their votes to the candidate in favour. The code snippet to implement this module is given below:

Code Snippet:

```
renderCandidates = (candidate) => {
    const castVote = async (id) => {
      await this.state.ElectionInstance.methods
        .vote(id)
        .send({ from: this.state.account, gas: 1000000 });
      window.location.reload();
    };
    const confirmVote = (id, header) => {
      var r = window.confirm(
        "Vote for " + header + "(" + candidate.slogan + ")" + ".\nAre you sure?"
      );
      if (r === true) {
        castVote(id);
      }
    };
    return (
      <div className="container-item">
        <div className="candidate-info">
          <h2>
```

```
            {(parseInt(candidate.id) + 1).toString()}) {candidate.header}
          </h2>
          <p className="slogan">{candidate.slogan}</p>
        </div>
        <div className="vote-btn-container">
          <button
            onClick={() => confirmVote(candidate.id, candidate.header)}
            className="vote-bth"
            disabled={
              !this.state.currentVoter.isRegistered ||
              !this.state.currentVoter.isVerified ||
              this.state.currentVoter.hasVoted
            }
          >
            Vote
          </button>
        </div>
      </div>
    );
  };
```

5.1.5 Module 5: Display Results

This module displays the results after the election has been ended. The winning candidate is declared in this module. The code snippet to implement this module is given below:

Code Snippet:

```
function displayWinner(candidates) {
  const getWinner = (candidates) => {
    // Returns an object having maxium vote count
    let maxVoteRecived = 0;
    let winnerCandidate = [];
    for (let i = 0; i < candidates.length; i++) {
      if (candidates[i].voteCount > maxVoteRecived) {
        maxVoteRecived = candidates[i].voteCount;
        winnerCandidate = [candidates[i]];
      } else if (candidates[i].voteCount === maxVoteRecived) {
        winnerCandidate.push(candidates[i]);
      }
    }
    return winnerCandidate;
  };
  const renderWinner = (winner) => {
    return (
      <div className="container-winner">
        <div className="winner-info">
          <p className="winner-tag">Most voted candidate</p>
          <h2> {winner.header}</h2>
          <p className="winner-slogan">{winner.slogan}</p>
        </div>
        <div className="winner-votes">
          <div className="votes-tag">Total Votes: </div>
```

```
          <div className="vote-count">{winner.voteCount}</div>
        </div></div>
    );
  };
  const winnerCandidate = getWinner(candidates);
  return <>{winnerCandidate.map(renderWinner)}</>;
}
export function displayResults(candidates) {
  const renderResults = (candidate) => {
    return (
      <tr>
        <td>{(parseInt(candidate.id) + 1).toString()}</td>
        <td>{candidate.header}</td>
        <td>{candidate.slogan}</td>
        <td>{candidate.voteCount}</td>
      </tr>
    );
  };
  return (
    <>
      <div className="container-item" style={{ border: "1px solid black" }}>
        <center>Final Results</center>
      </div>
      {candidates.length > 0 ? (
        <div className="container-main">{displayWinner(candidates)}</div>
      ) : null}
      <div className="container-main" style={{ borderTop: "1px solid" }}>
        <h2>Results</h2>
        <small>Total candidates: {candidates.length}</small>
        {candidates.length < 1 ? (
          <div className="container-item attention">
            <center>No candidates.</center>
          </div>
        ) : (
          <>
            <div className="container-item">
              <table>
                <tr>
                  <th>Id</th>
                  <th>Candidate</th>
                  <th>Party</th>
                  <th>Votes</th>
                </tr>
                {candidates.map(renderResults)}
              </table>
            </div>
          </>
        )}
      </div>
    </>
  );
}
```

5.2 Smart Contracts

These are logic that gets executed when a condition is fulfilled. Smart contracts are coded using solidity programming language. The code snippet to implement this election smart contract is:

Code Snippet:

```
contract Election {
    address public admin;
    uint256 candidateCount; uint256 voterCount;
    bool start; bool end;
    constructor() public { // Initilizing default values
        admin = msg.sender;
        candidateCount = 0;
        voterCount = 0;
        start = false; end = false;
    }

    function getAdmin() public view returns (address) { return admin; }
    modifier onlyAdmin() { require(msg.sender == admin); }

 struct Candidate {
        uint256 candidateId;
        string header; string slogan;
        uint256 voteCount;
    }
    mapping(uint256 => Candidate) public candidateDetails;

    // Adding new candidates
    function addCandidate(string memory _header, string memory _slogan)
        public
        onlyAdmin {
        Candidate memory newCandidate =
            Candidate({
                candidateId: candidateCount, header: _header,
                slogan: _slogan, voteCount: 0
            });
        candidateDetails[candidateCount] = newCandidate;
        candidateCount += 1;
    }

    struct ElectionDetails { // Modeling a Election Details
        string adminName; string adminEmail; string adminTitle;
        string electionTitle; string organizationTitle;
    }
    ElectionDetails electionDetails;

    function setElectionDetails(
        string memory _adminName, string memory _adminEmail,
        string memory _adminTitle,
        string memory _electionTitle, string memory _organizationTitle
    )
        public
```

```
        onlyAdmin {
        electionDetails = ElectionDetails( _adminName, _adminEmail,
            _adminTitle, _electionTitle, _organizationTitle );
        start = true;
        end = false;
    }

    function getAdminName() public view returns (string memory) {
        return electionDetails.adminName;
    }
    function getAdminEmail() public view returns (string memory) {
        return electionDetails.adminEmail;
    }
    function getAdminTitle() public view returns (string memory) {
        return electionDetails.adminTitle;
    }
    function getElectionTitle() public view returns (string memory) {
        return electionDetails.electionTitle;
    }
    function getOrganizationTitle() public view returns (string memory) {
        return electionDetails.organizationTitle;
    }
    function getTotalCandidate() public view returns (uint256) {
        return candidateCount; // Get candidates count
    }
    function getTotalVoter() public view returns (uint256) {
        return voterCount; // Get voters count
    }

    struct Voter { // Modeling a voter
        address voterAddress;
        string name;
        string phone;
        bool isVerified;
        bool hasVoted;
        bool isRegistered;
    }
    address[] public voters; // Array of address to store address of voters
    mapping(address => Voter) public voterDetails;

    // Request to be added as voter
    function registerAsVoter(string memory _name, string memory _phone) public {
        Voter memory newVoter =
            Voter({
                voterAddress: msg.sender,
                name: _name,
                phone: _phone,
                hasVoted: false,
                isVerified: false,
                isRegistered: true
            });
        voterDetails[msg.sender] = newVoter;
```

```
        voters.push(msg.sender);
        voterCount += 1;
    }

    // Verify voter
    function verifyVoter(bool _verifedStatus, address voterAddress)
        public
        // Only admin can verify
        onlyAdmin {
        voterDetails[voterAddress].isVerified = _verifedStatus;
    }

    function vote(uint256 candidateId) public { // Vote
        require(voterDetails[msg.sender].hasVoted == false);
        require(voterDetails[msg.sender].isVerified == true);
        require(start == true);
        require(end == false);
        candidateDetails[candidateId].voteCount += 1;
        voterDetails[msg.sender].hasVoted = true;
    }

    function endElection() public onlyAdmin { // End election
        end = true;
        start = false;
    }

    function getStart() public view returns (bool) {
        return start;
    }

    function getEnd() public view returns (bool) {
        return end;
    }
}
```

CHAPTER 6

SNAPSHOTS

This chapter shows some of the working snapshots of our project “Voting Application using Blockchain Technology”.

6.1 Creating the Blockchain Network

Before starting to work with our application, we need to first create the block chain network. To do this, we have to migrate to the project folder in the command prompt and run the command ganache-cli in the command prompt. On doing so, we get 10 accounts created by default by ethereum blockchain environment with the first account corresponding to the admin and the remaining nine accounts corresponding to voters as shown in figure 6.1.

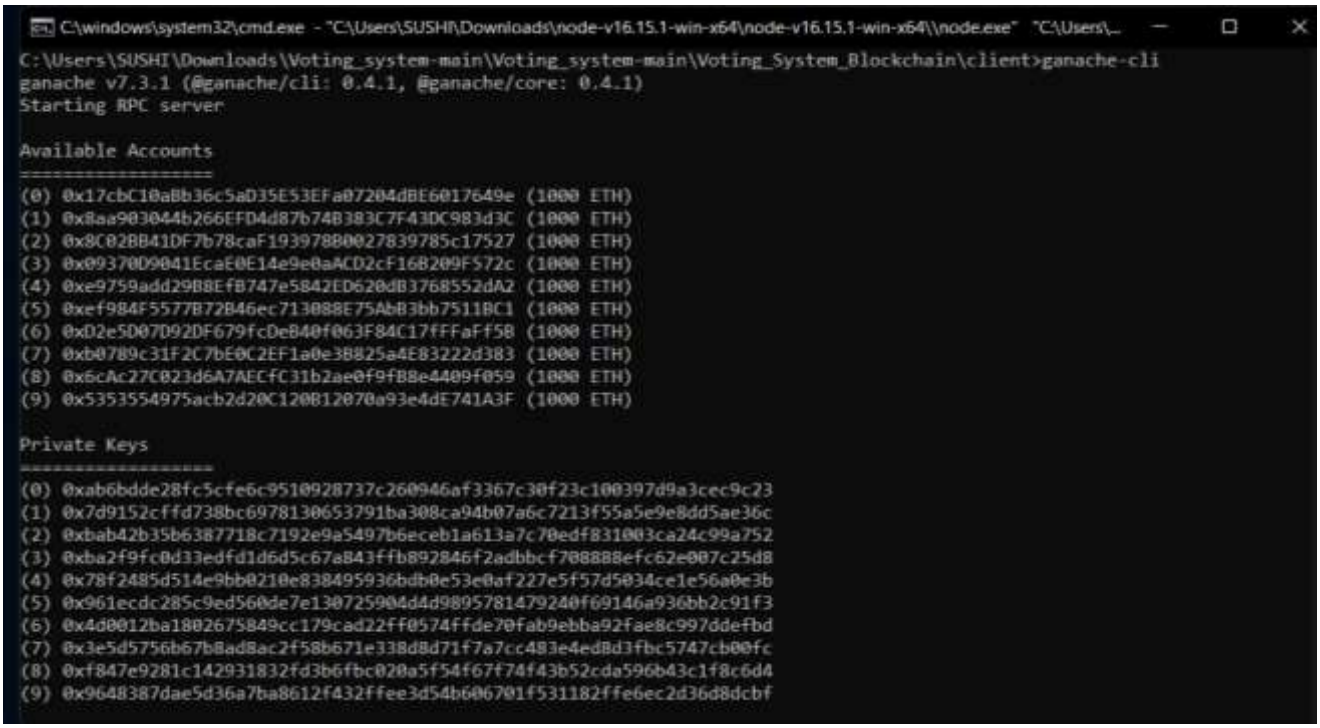

Fig 6.1: Creating Blockchain Network

6.2 Deploying Smart Contracts

After the block chain network is created, we need to deploy the smart contracts. To do so, we need to open another command prompt and migrate to the project folder. After this, we have to run the command truffle migrate in the command prompt to ensure that the smart contract is deployed as shown in figure 6.2.

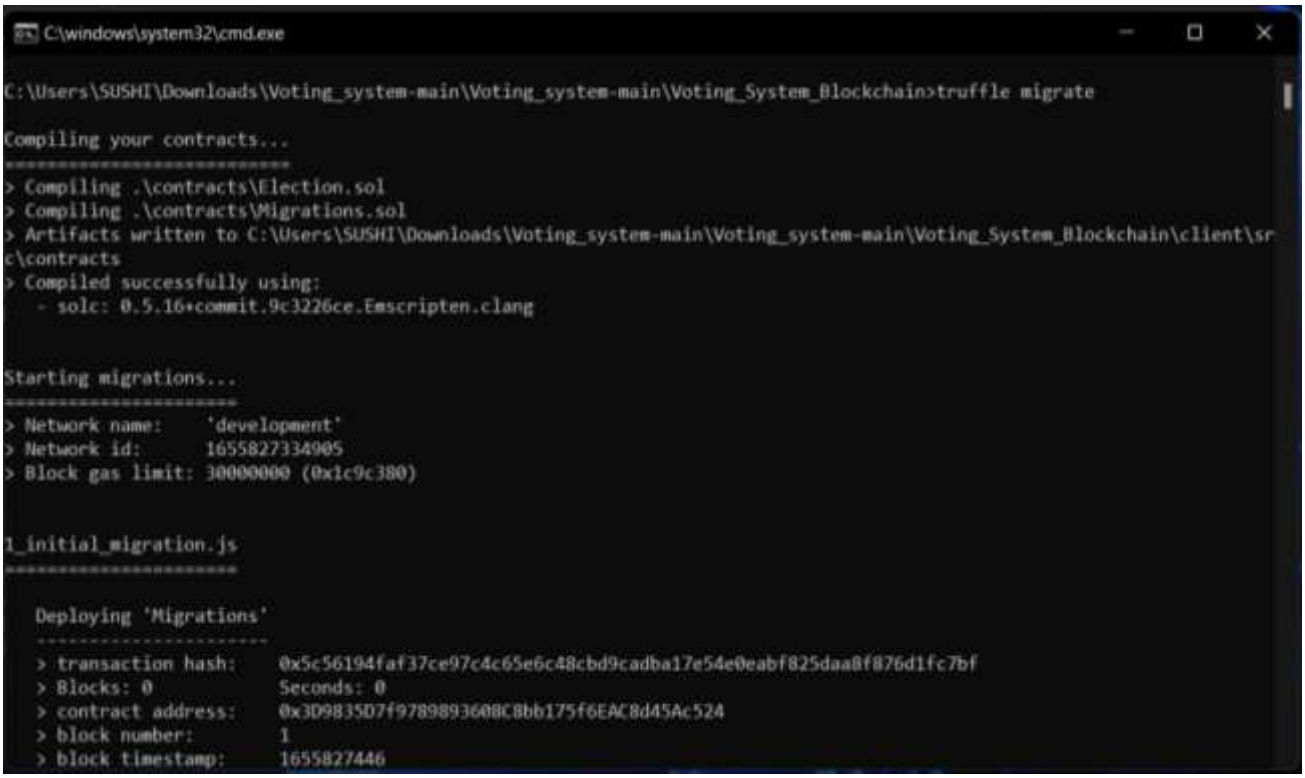

Fig 6.2: Deploying Smart Contracts

6.3 Starting the Application

After the creation of block chain network and deployment of smart contracts, we need to start the application by starting the react server. This is done by opening the command prompt, migrating to the project folder and then running the command cd client followed by npm start as shown in figure 6.3.

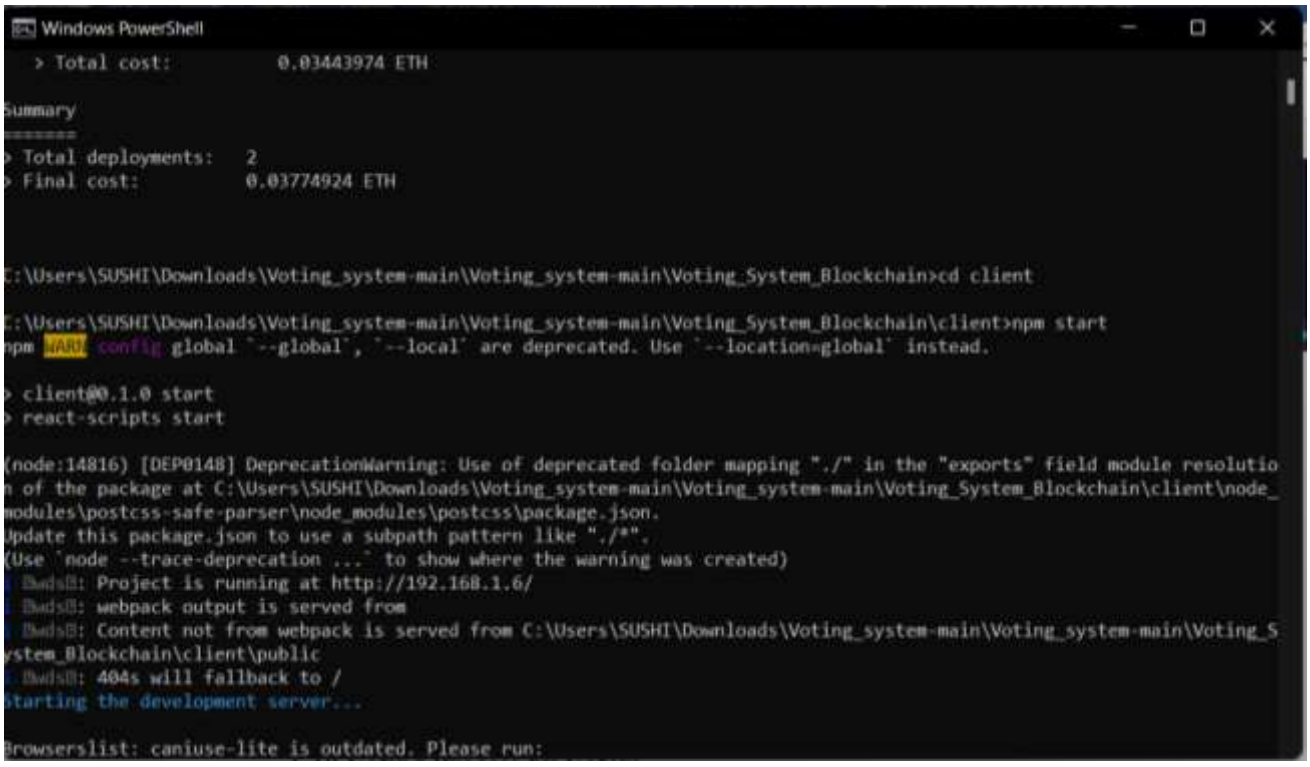

Fig 6.3: Starting the application

6.4 Admin Registration to Conduct Election

When the application starts, the first account (Admin) is imported to the application through the metamask extension and the admin has to register with the system in order to initiate the voting activity. This can be seen in figure 6.4.

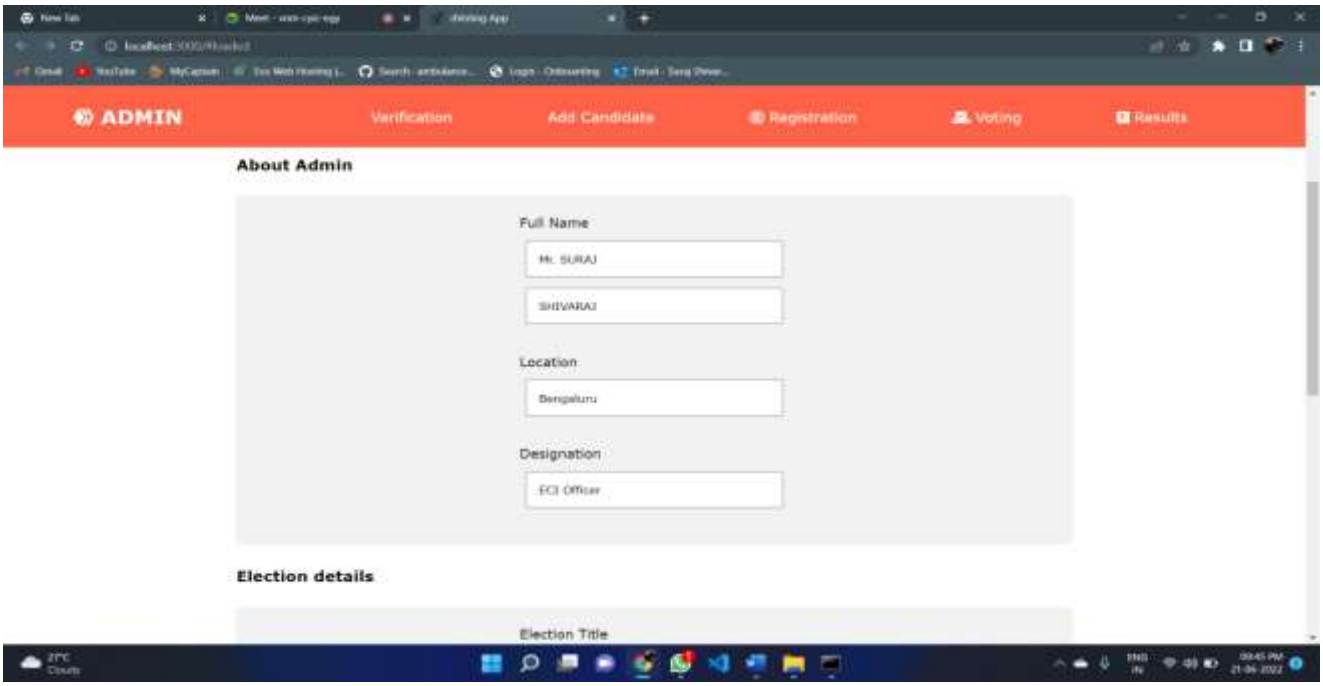

Fig 6.4: Admin Registration

6.5 Admin Adding Candidates

After the admin registration is complete. The admin can add candidates who wish to contest in the elections by providing the candidate name and the party they belong to. This can be seen in figure 6.5.

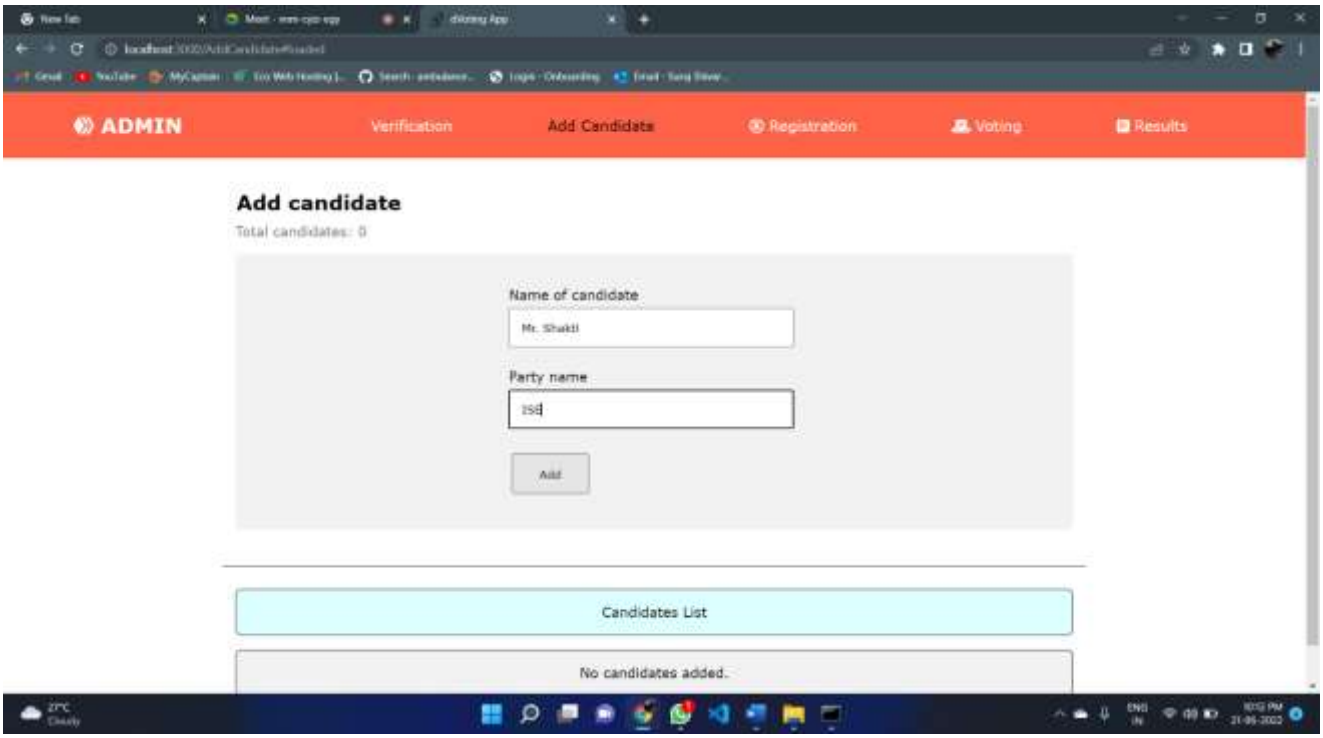

Fig 6.5: Admin Adding Candidates

After the admin adds the interested candidates, they are added to the candidate list. This can be seen in figure 6.6.

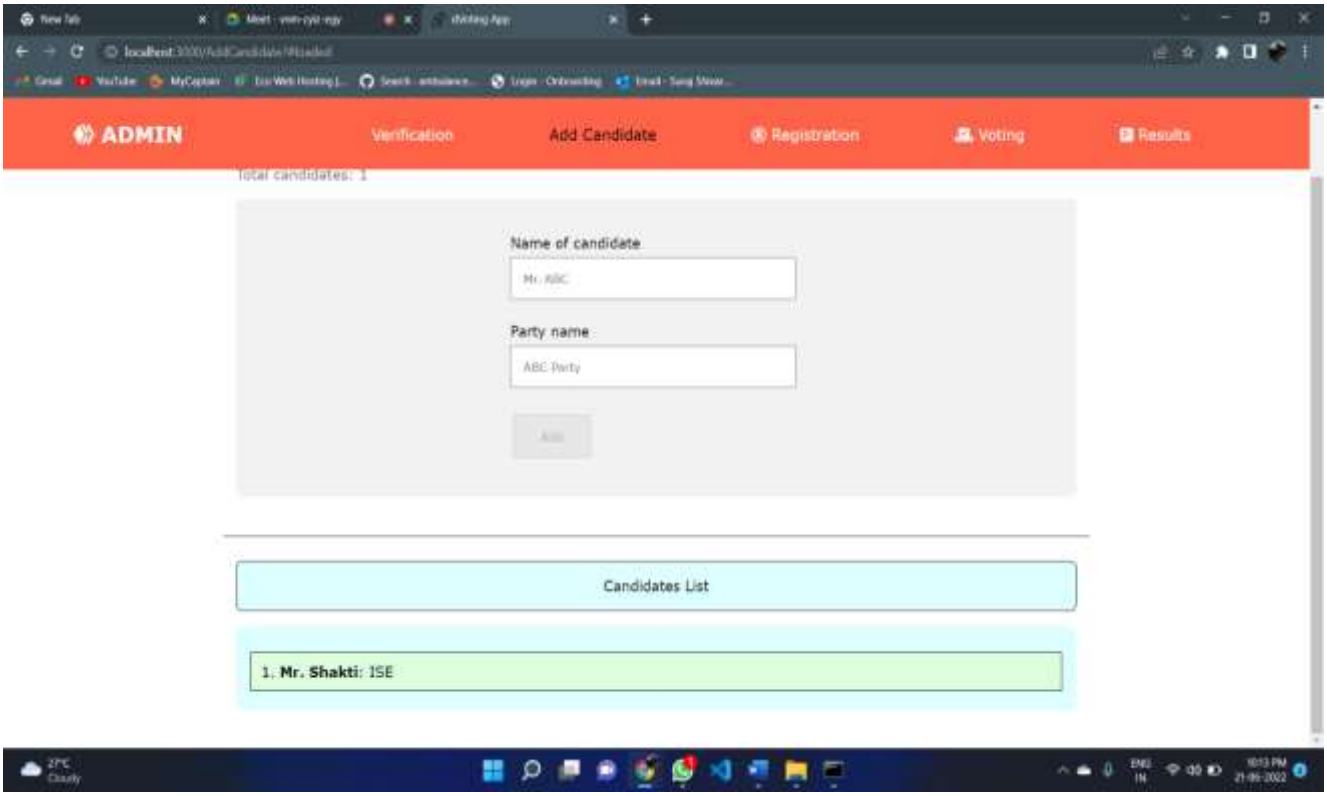

Fig 6.6: Candidate List after Addition

6.6 Voter Registration to Vote

Here, the voter is provided with an interface, to register themselves with the voting application in order to take part in the voting activity. This can be seen in figure 6.7.

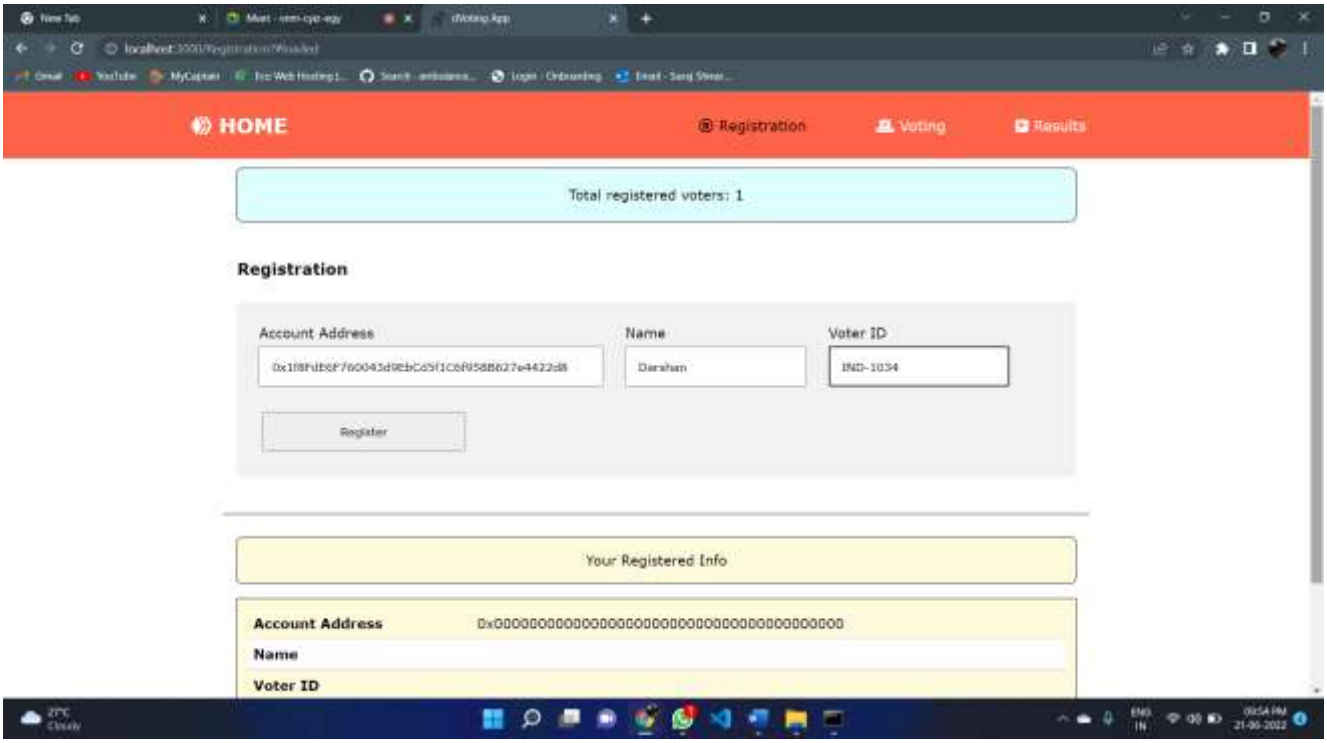

Fig 6.7: Voter Registration

6.7 Admin Verification to Approve Voters

Here, the admin verifies the details given by voters before approving them. This can be seen in the figure 6.8.

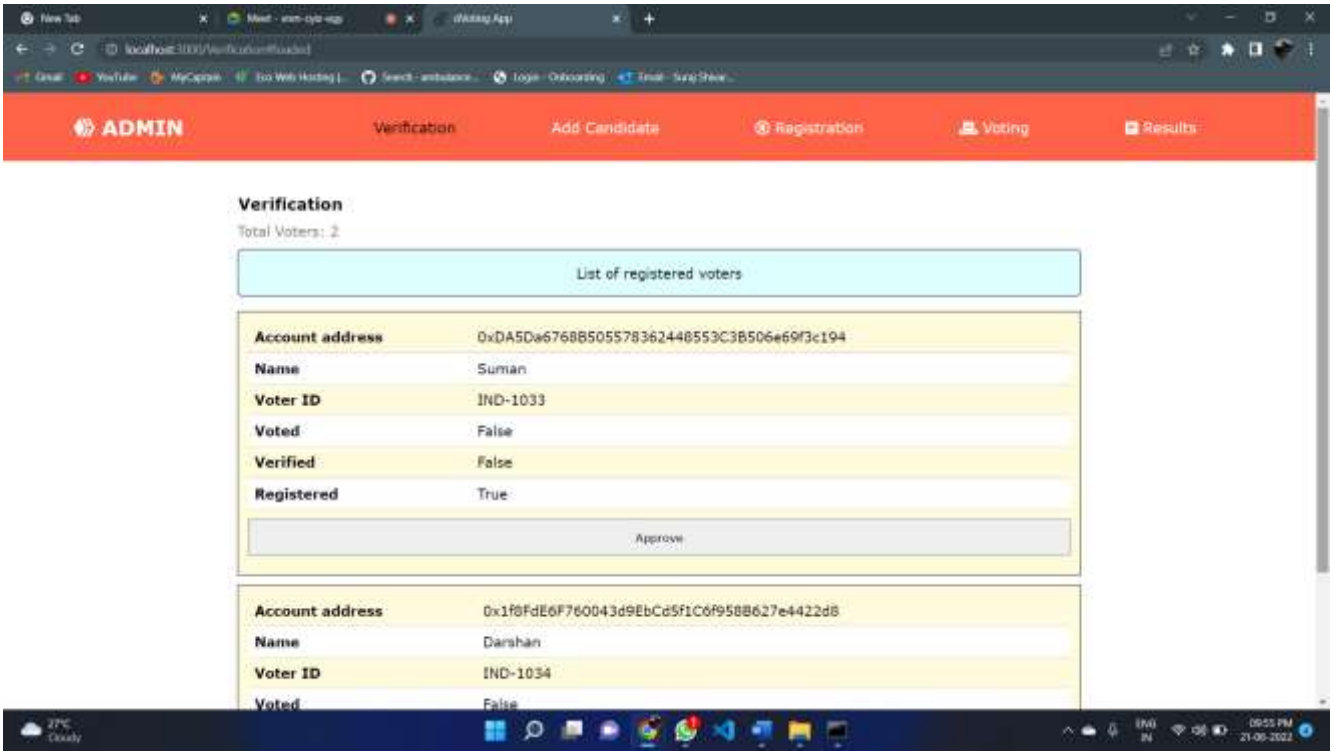

Fig 6.8: Admin Verification of Voters

After the admin approves the voters, they are eligible to cast their vote. Since the voter has not voted yet the value of Voted is set to False. This can be seen in figure 6.9.

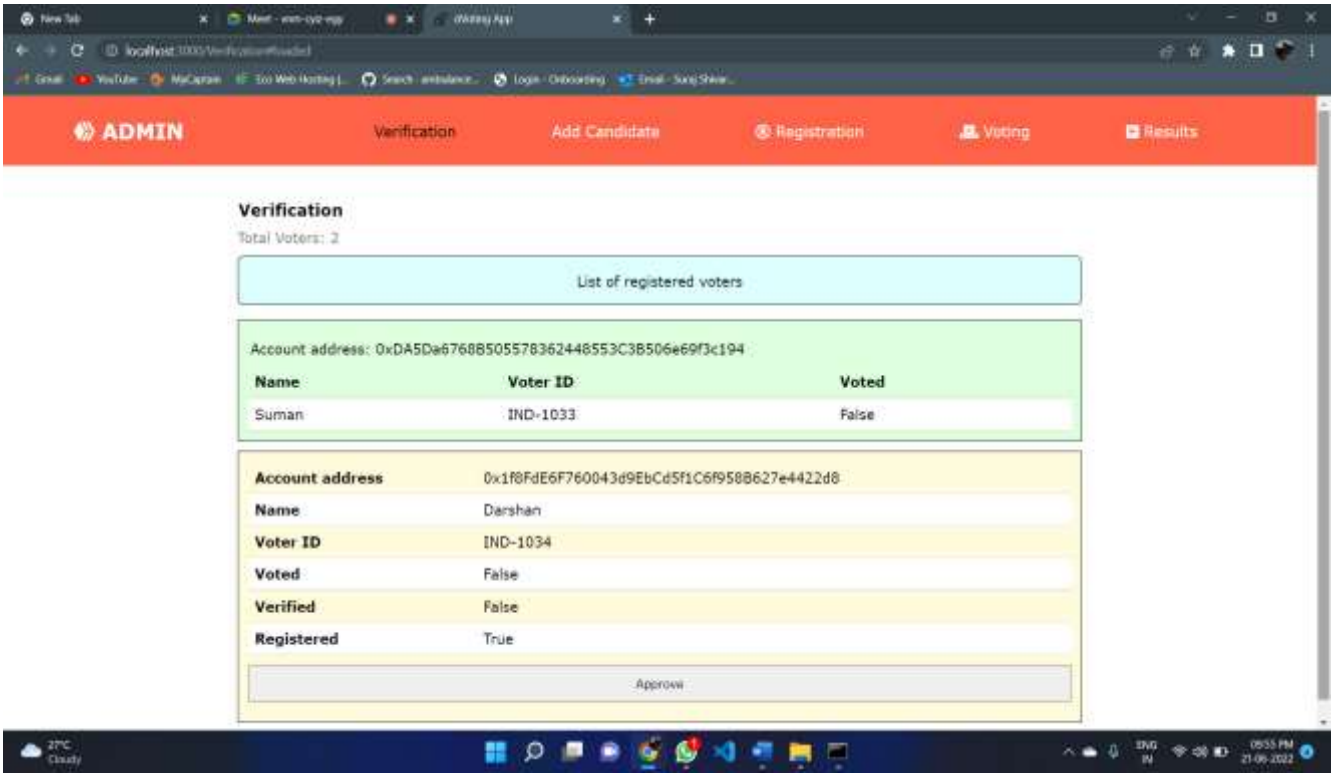

Fig 6.9: Voter Approved by Admin

6.8 Starting the Election

Once all the candidates are added and the voters are approved, the admins begins the voting process. This is done by clicking on the start button as shown in figure 6.10.

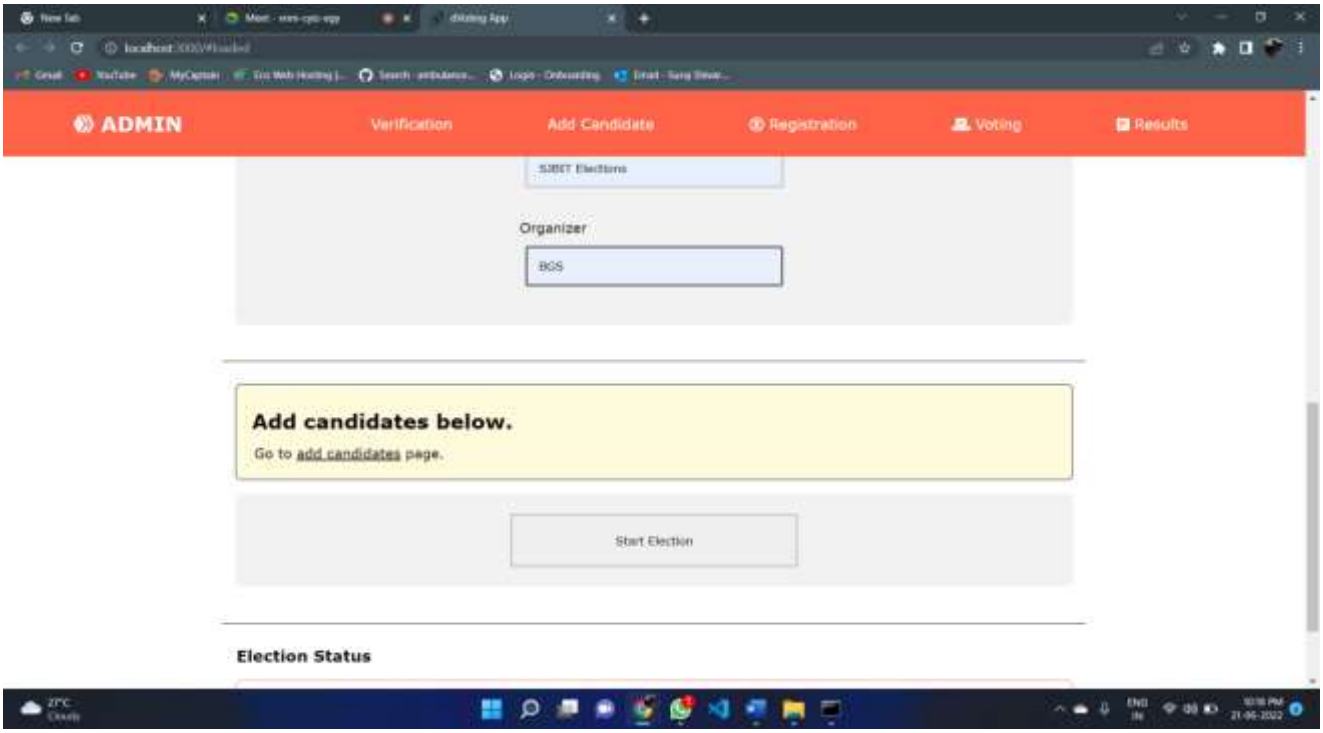

Fig 6.10: Admin Starting Election

6.9 Voting Process by Voters

Here, only the voters approved by the admin will be able to cast their vote for the desired candidate. This can be seen in figure 6.11.

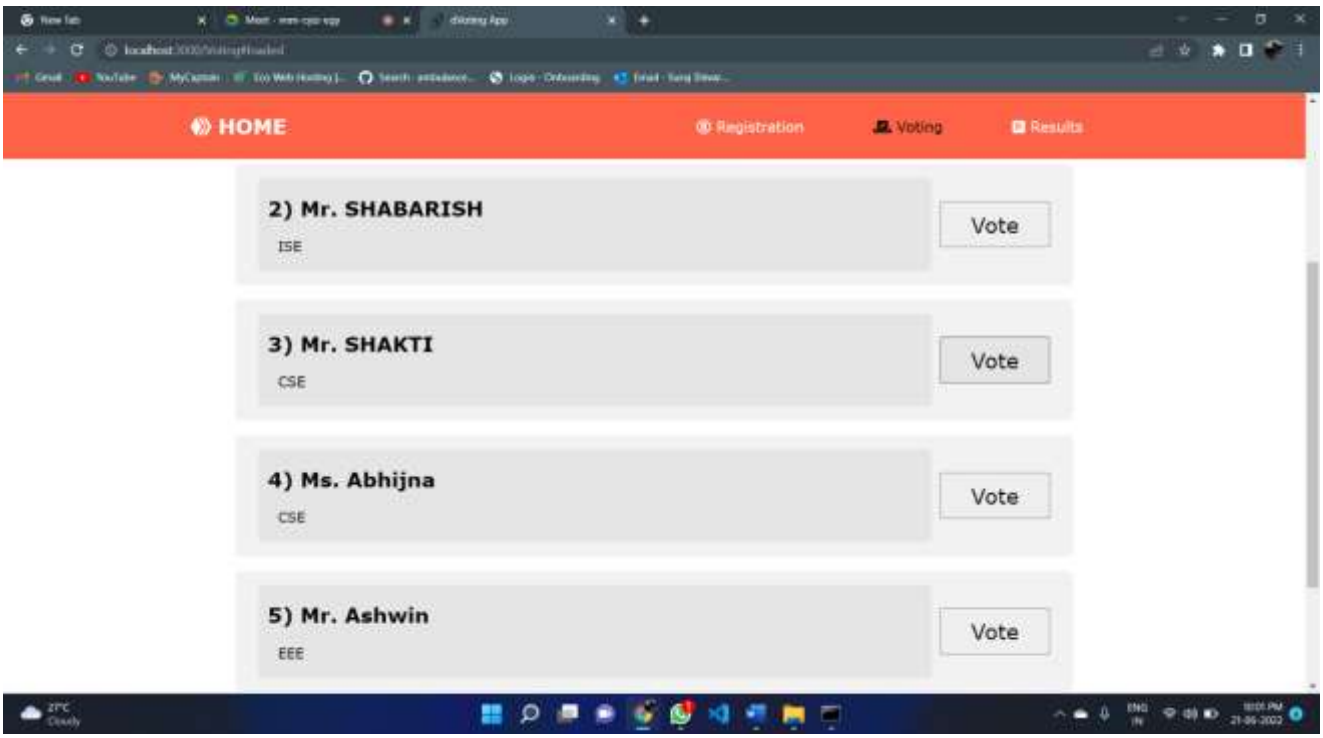

Fig 6.11: Casting Votes

After the vote button against the desired candidate is clicked, a confirmation is prompted to the voter. Upon clicking yes, the vote is recorded in the blockchain. Clicking cancel will not record the vote casted. This can be seen in figure 6.12.

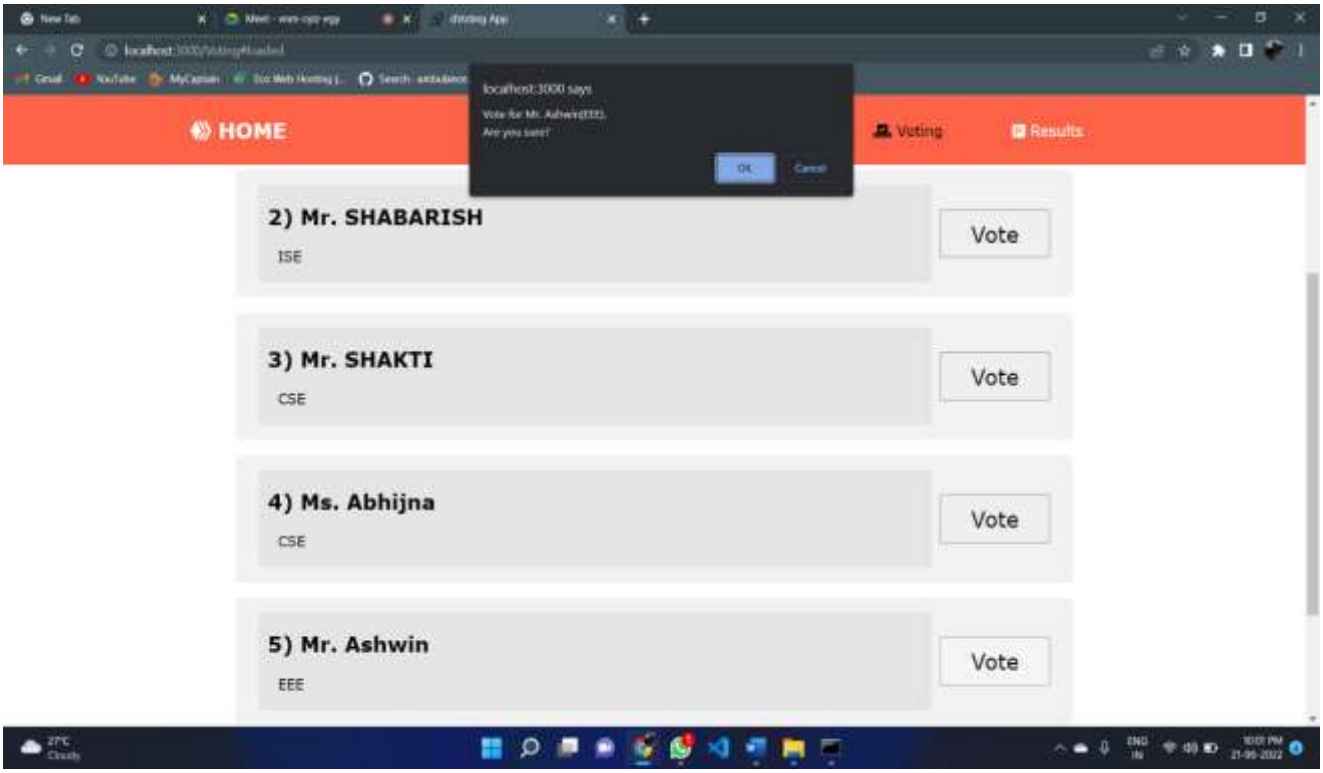

Fig 6.12: Vote Confirmation

Figure 6.13 shows the results of after confirming a vote to a candidate.

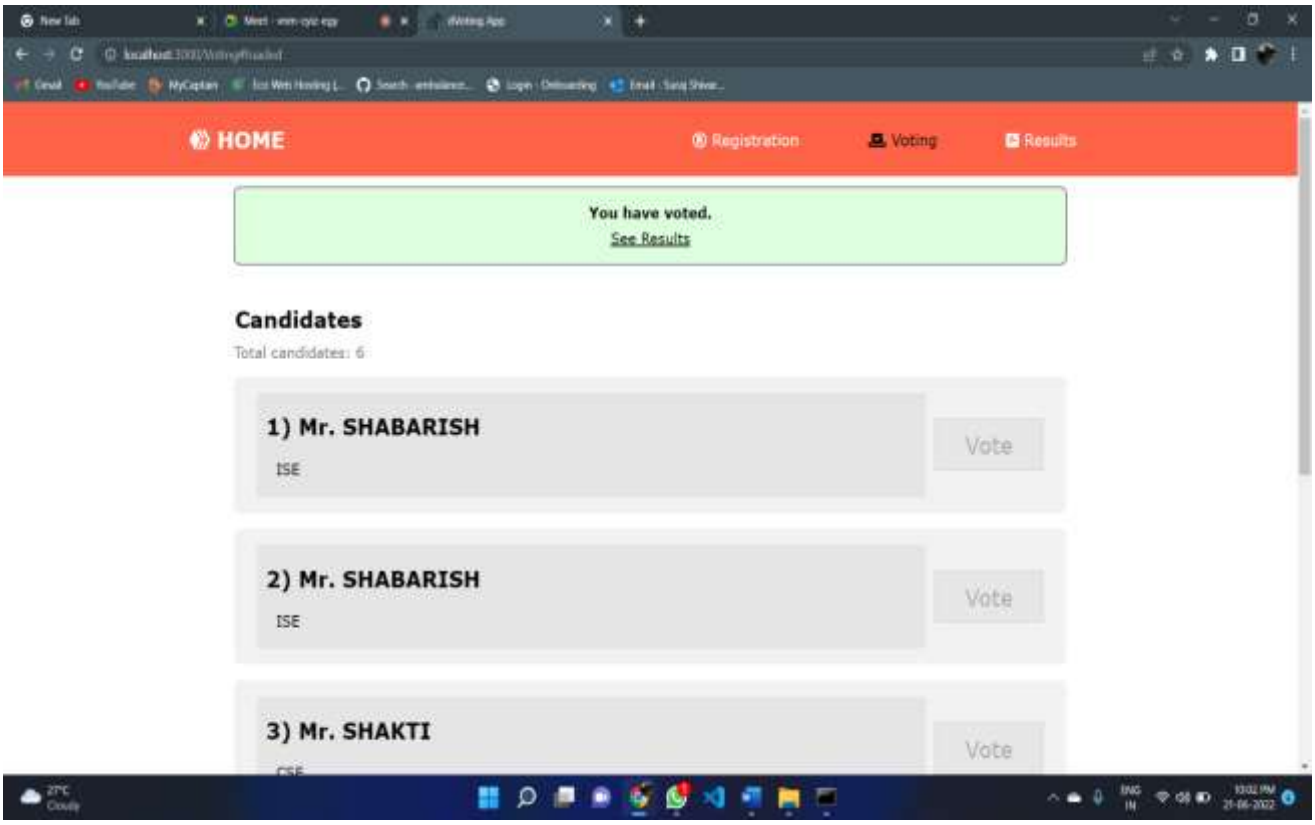

Fig 6.13: After Voting a Candidate

If the voter clicks on see results as shown in figure 6.13, the system will prompt a message as shown in figure 6.14.

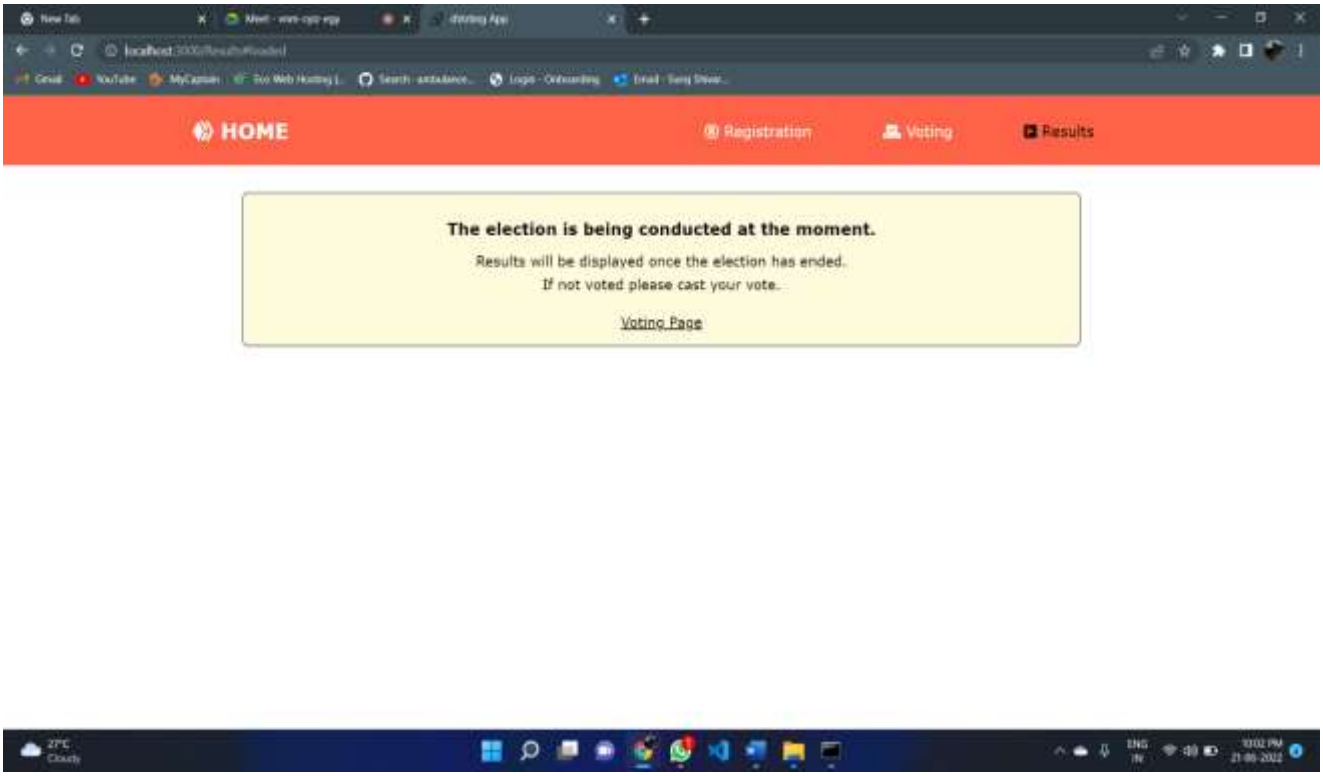

Fig 6.14: Message to Voters before Election End

6.10 Ending the Election

After the election interval is lapsed, the admin will end the election. This is done by clicking on the button End election. Now, no voters will be able to cast their votes. This can be seen in figure 6.15.

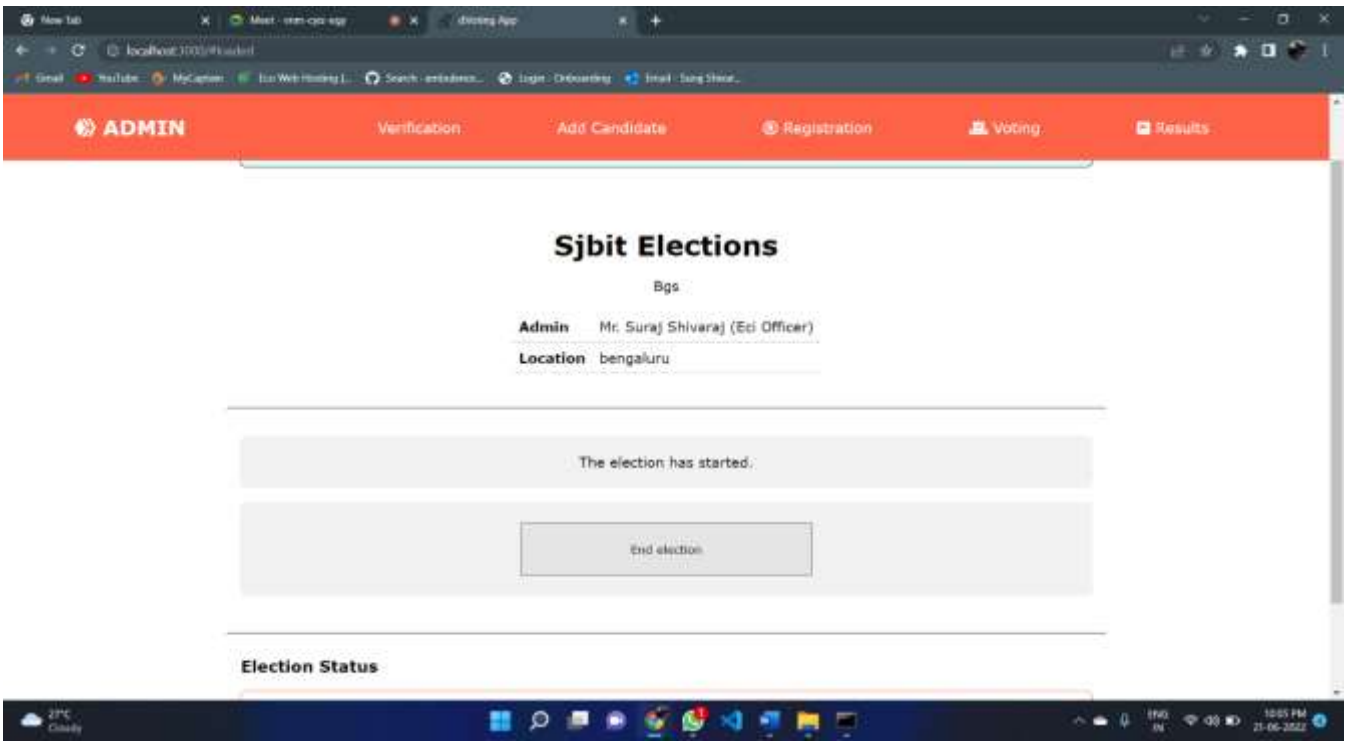

Fig 6.15: Admin Ending Election

6.11 Publishing Results

Here, the final results of the election are published and the winner is declared by the system. This can be seen in figure 6.16.

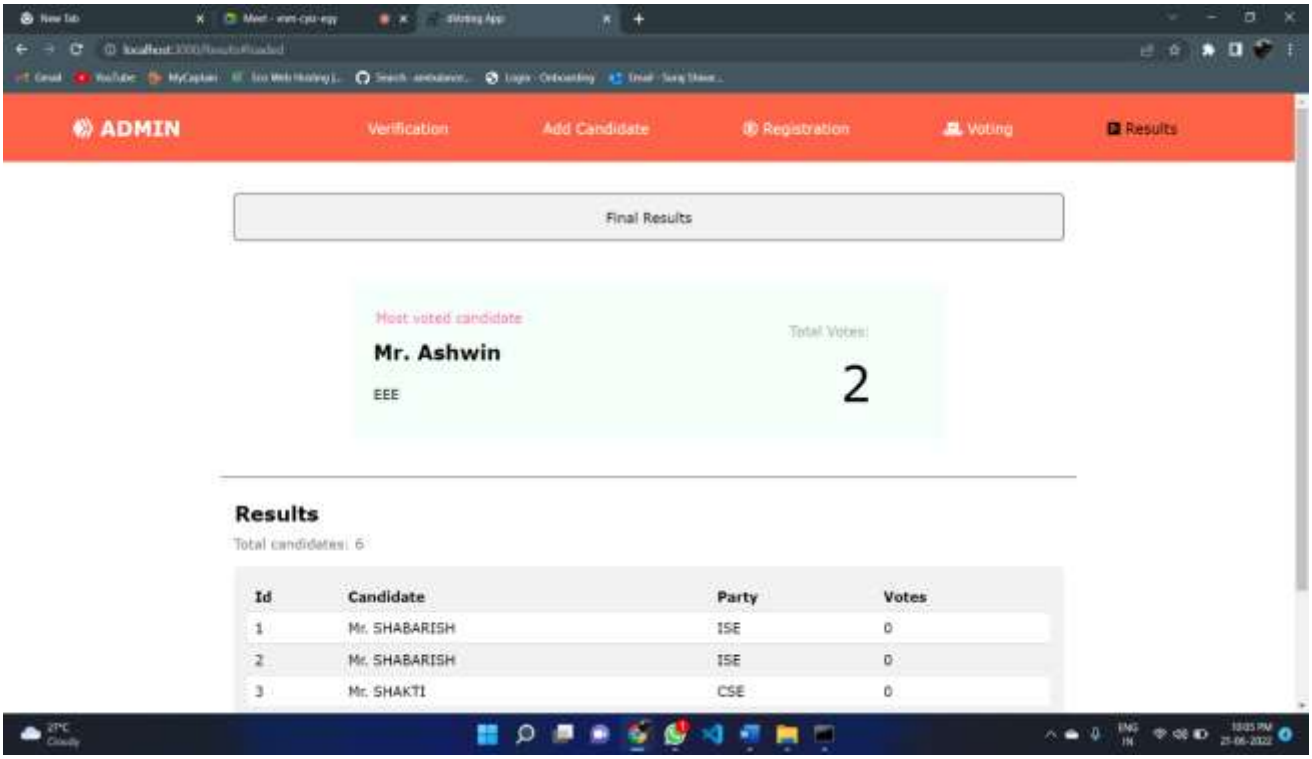

Fig 6.16: Publishing results

Also the details regarding the number of voted casted for other candidate can be retrieved as shown in figure 6.17.

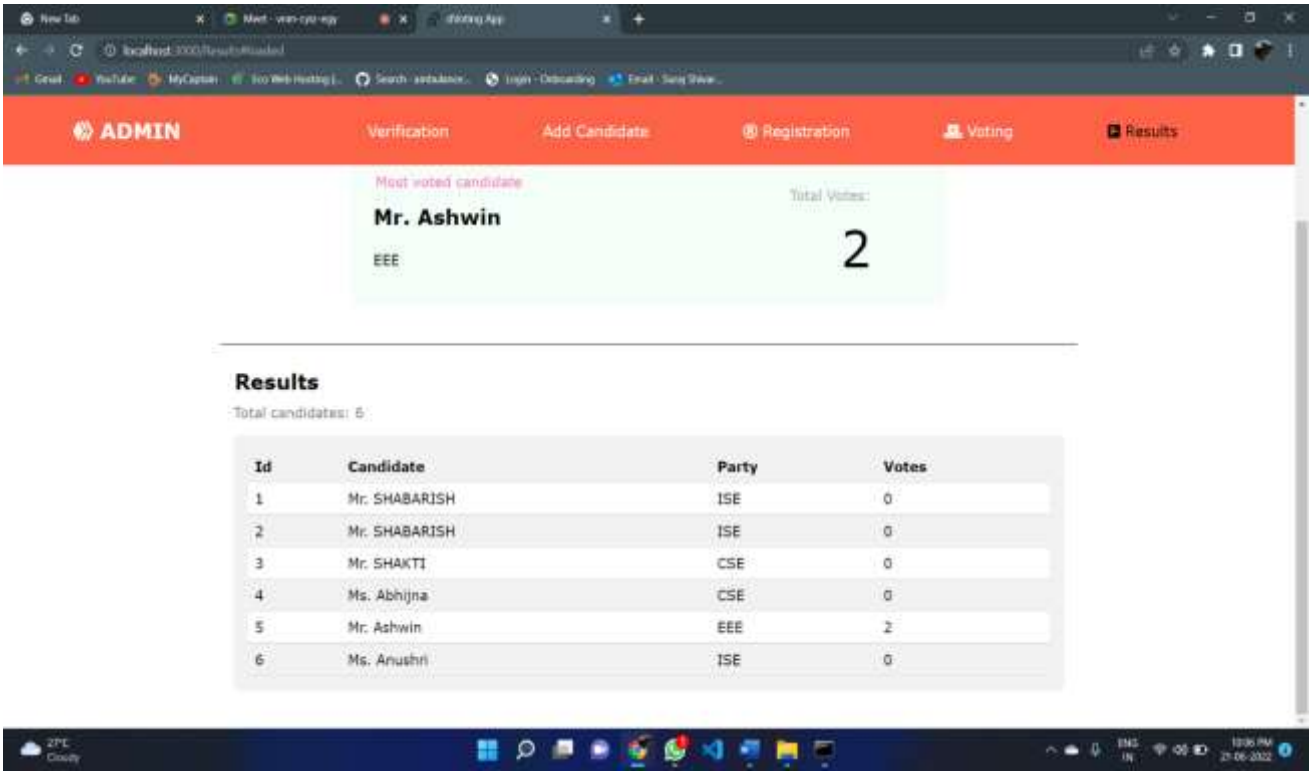

Fig 6.17: Other Candidate Results

CONCLUSION

The purpose of developing a blockchain-based solution for the voting system is to build trust among voters and make them believe that their voting integrity is kept safe. The amount of money spent on voting activity in any country is very high for the traditional voting system, whereas the proposed system of using the blockchain voting systems makes voting process cheaper, faster, transparent and trustworthy.

In our project, we have developed a transparent, secure and digital voting system using smart contracts and block chain technology that will be able to overcome the limitations seen in traditional method as well as in other digital voting systems.

Applications:

- Our application can be used in schools or colleges in electing a student for the post of president of student union or student alumni.
- It can also be used in small organisations or community to formally elect a candidate.

As a future work we can try to add a system that will be able to generate reports of end results. At the same time, adopt biometric based authentication for voter verification. Also, we can try to create multiple instances rather than redeploying the smart contracts again and again after one election instance is concluded.

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